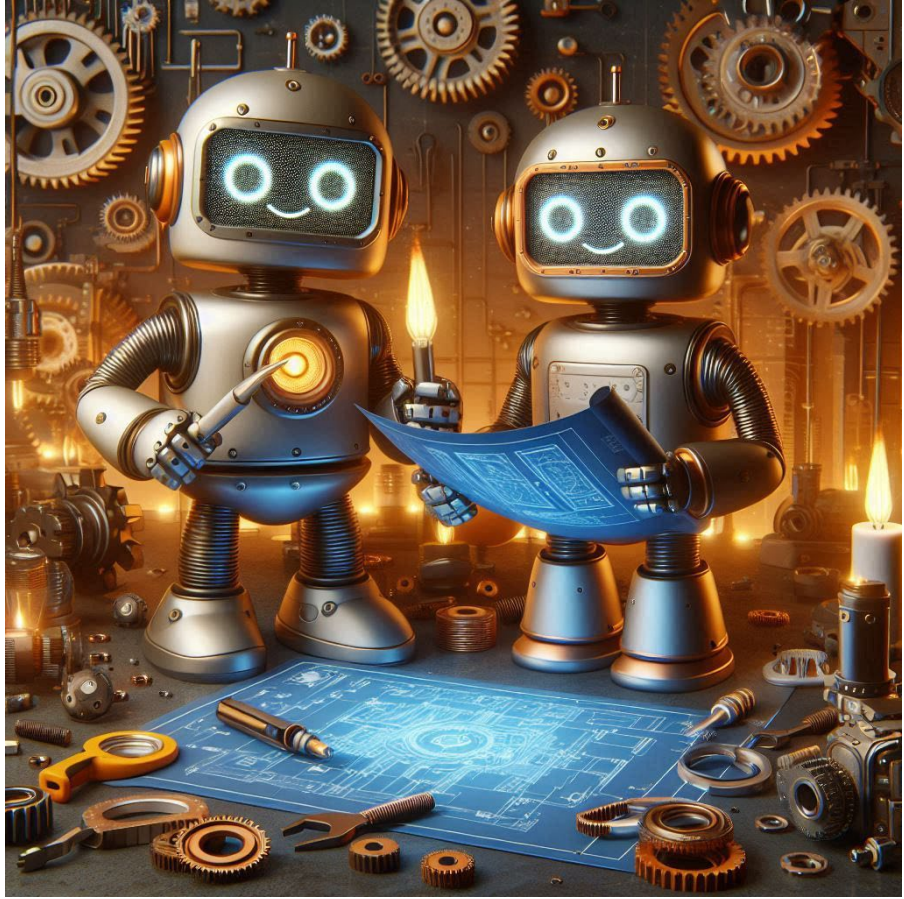
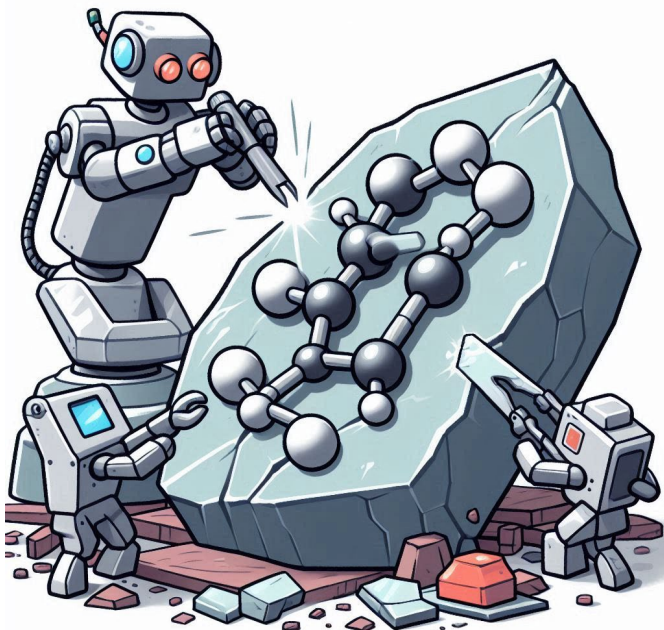




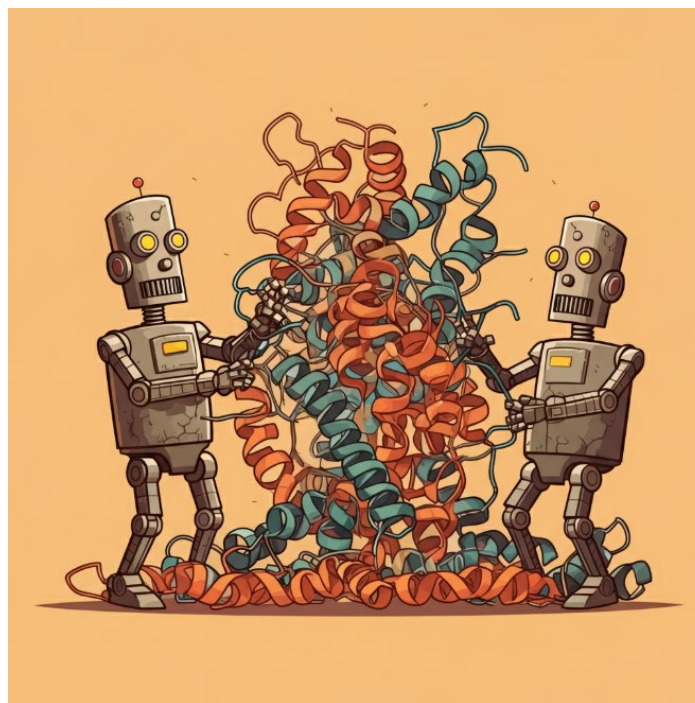
AI in Drug Discovery Workshop

Session 3 – Practical Applications

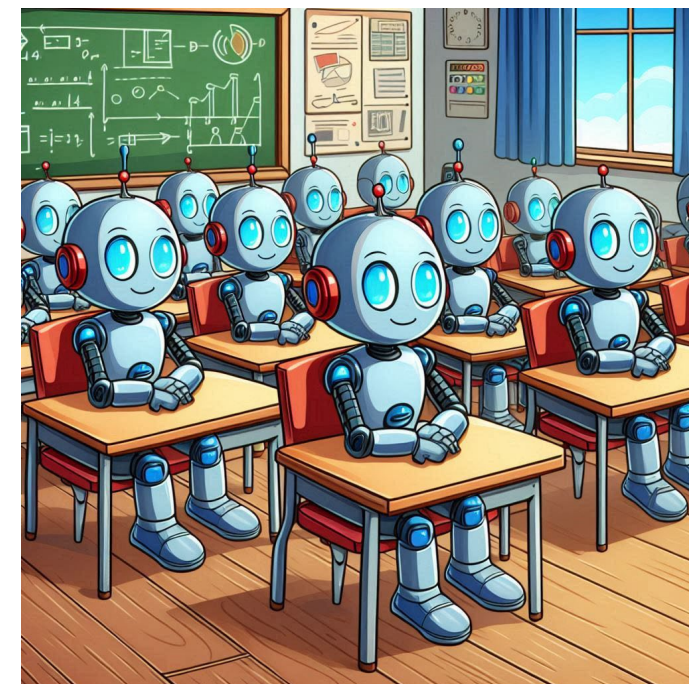
Practical Applications of Machine Learning in Drug Discovery



Molecule Generation

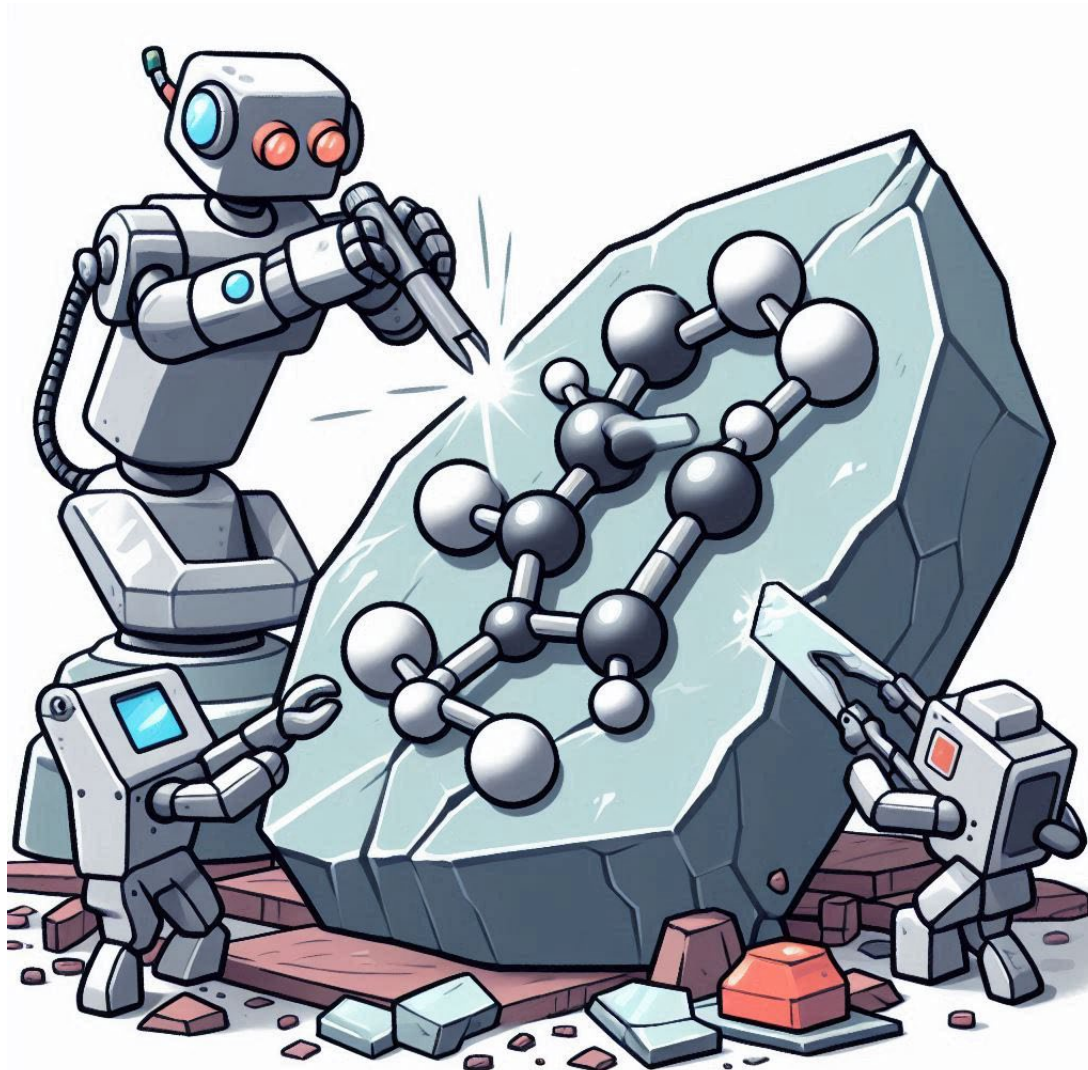


Protein Structure Prediction



Active Learning

Molecule Generation



Large Language Models (LLMs) Learn Next Token Probabilities

Given the phrase “The cat is”, what is the next token?

In this case, a token is a word

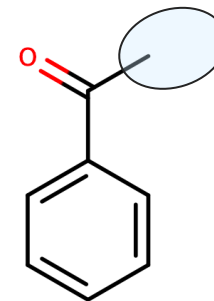
$$P(\textit{blue}|\textit{The, cat, is}) = .1$$

$$P(\textit{black}|\textit{The, cat, is}) = .85$$

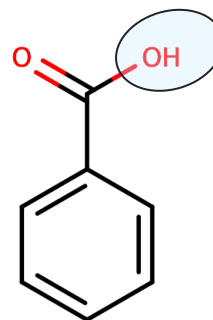
$$P(\textit{green}|\textit{The, cat, is}) = .05$$

Next Token Probabilities Can Also Be Applied to Chemistry

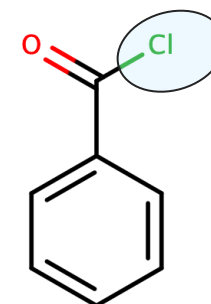
$$P(C, c1ccccc1C(=O)) = 0.2$$



$$P(O, c1ccccc1C(=O)) = 0.75$$



$$P(Cl, c1ccccc1C(=O)) = 0.05$$



Training a Recurrent Neural Network (RNN) to Generate “Shakespeare”

Prior

First Citizen:
Before we proceed any further, hear me speak.

All:
Speak, speak.

First Citizen:
You are all resolved rather to die than to famish?

All:
Resolved. resolved.

First Citizen:
First, you know Caius Marcius is chief enemy to the people.

All:
We know't, we know't.

First Citizen:
Let us kill him, and we'll have corn at our own price.
Is't a verdict?

All:
No more talking on't; let it be done: away, away!

Second Citizen:
One word, good citizens.

First Citizen:
We are accounted poor

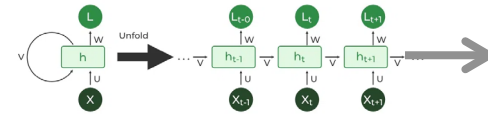
etc.

Embedding

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```

etc.

RNN



Output

ROMEO:
It mares to brave, my lord: good peace! Feeping to be crown'd,
I dreamt occapity: a baudish, make thee mellow.

VINCENTIO:
I' the duke, sir, their beaming on a despite of more
view to the prozont; and my vow,
Tuguse, when dreadful Norts of Hills,
Unless you be the moonest malice or tent to false;
whistle and hours are too much commended.

JULIET:
Dost thou not, but have they may come there.

ESCALUS:
We have: I'll visit with thee
And bid Bimine own fight and beggarly; whom
Too little liking, for I know thou art not vice,
Though father is no less a man that left your high
As twenty thousand end,
Wrangening so well as eat him by changing
Against the Duke of York, Summon are me:
I am she, poor soul, I envy nothing feast,
Alack the heaven gails of mine heir toes
was in the view of Juliet o' the seas;
And every gash of invitation, but not
His hands received the estremest pass
And safely with my hands.

etc.

Training a Recurrent Neural Network (RNN) to Generate “Shakespeare”

Prior

First Citizen:
Before we proceed any further, hear me speak.

All:
Speak, speak.

First Citizen:
You are all resolved rather to die than to famis

All:
Resolved. resolved.

First Citizen:
First, you know Caius Marcius is chief enemy to

All:
We know't, we know't.

First Citizen:
Let us kill him, and we'll have corn at our own
Is't a verdict?

All:
No more talking on't; let it be done: away, away

Second Citizen:
One word, good citizens.

First Citizen:
We are accounted poor

etc.

ROMEO:
It mare to brave, my lord: good peace! Feeping to be crown'd,
I dreamt occapity: a baudish, make thee mellow.

VINCENTIO:
I' the duke, sir, their beaming on a despite of more
view to the prozont; and my vow,
Tuguse, when dreadful Norts of Hills,
Unless you be the moonest malice or tent to false;
whistle and hours are too much commended.

JULIET:
Dost thou not, but have they may come there.

ESCALUS:
We have: I'll visit with thee
And bid Bimine own fight and beggarly; whom
Too little liking, for I know thou art not vice,
Though father is no less a man that left your high
As twenty thousand end,
Wrangening so well as eat him by changing
Against the Duke of York, Summon are me:
I am she, poor soul, I envy nothing feast,
Alack the heaven gails of mine heir toes
was in the view of Juliet o' the seas;
And every gash of invition, but not
His hands received the estremerest pass
And safely with my hands.

Output

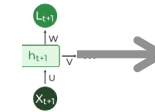
ROMEO:
It mare to brave, my lord: good peace! Feeping to be crown'd,
I dreamt occapity: a baudish, make thee mellow.

VINCENTIO:
I' the duke, sir, their beaming on a despite of more
view to the prozont; and my vow,
Tuguse, when dreadful Norts of Hills,
Unless you be the moonest malice or tent to false;
whistle and hours are too much commended.

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Alack the heaven gails of mine heir toes
was in the view of Juliet o' the seas;
And every gash of invition, but not
His hands received the estremerest pass
And safely with my hands.

etc.



Training a Recurrent Neural Network (RNN) to Generate SMILES

Prior

Embedding

RNN

Output

```

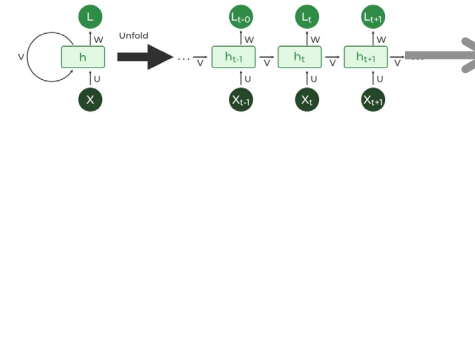
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etc.

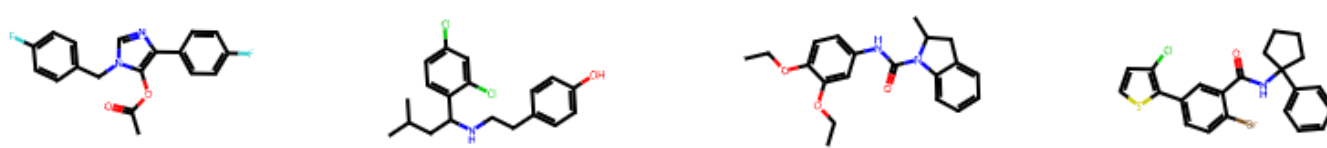
etc.

[illegible]

etc.

Training a Recurrent Neural Network (RNN) to Generate SMILES

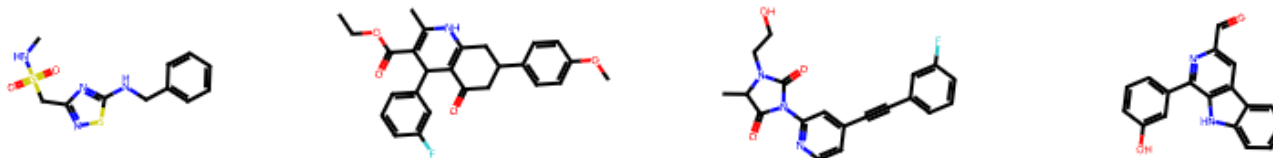
Prior

[illegible]

Output

[illegible]

etc.



etc.

Augmented Likelihood Incorporates Scoring Into Molecule Generation

Prior

Embedding

RNN

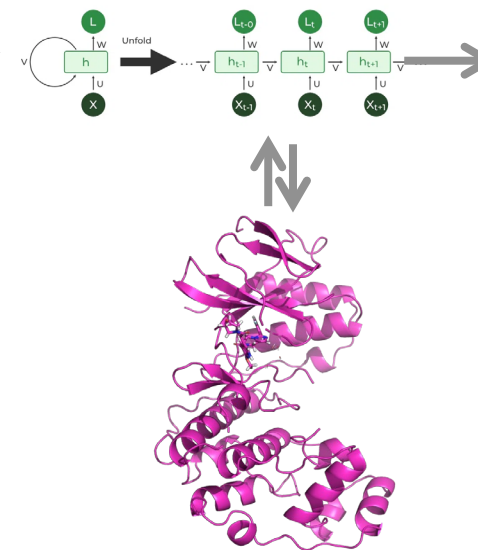
Output

[illegible]

etc.

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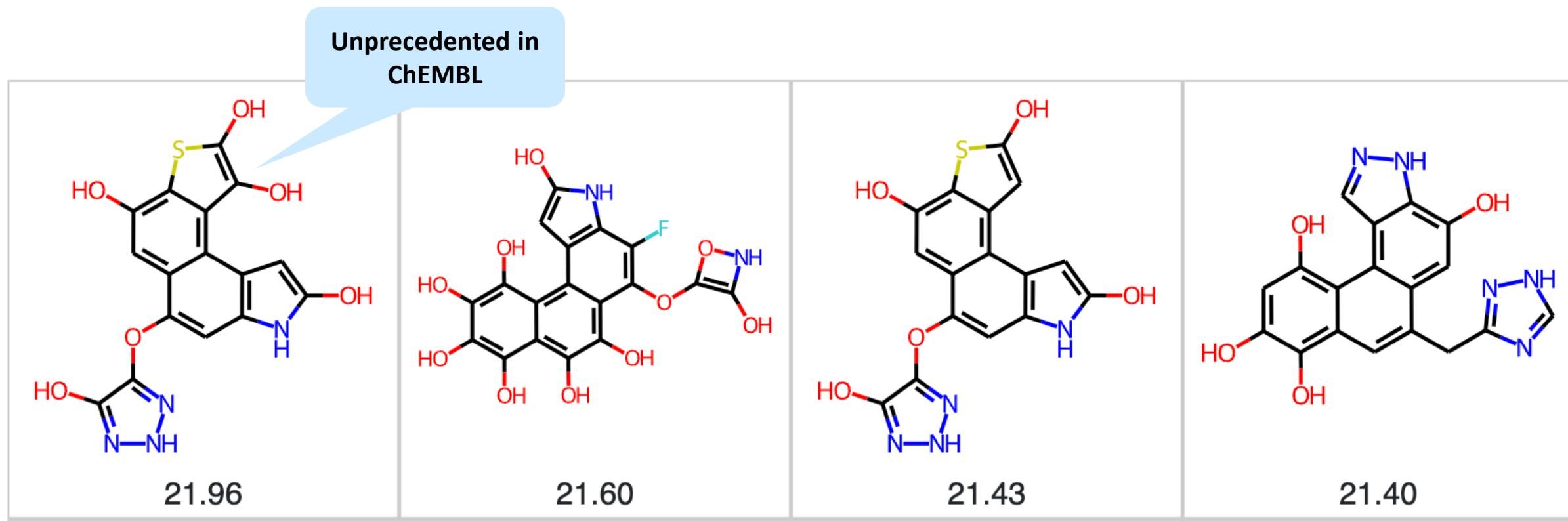
[illegible]

etc.

Augmented likelihood

$$\log P(A)_U = \log P(A)_{prior} + \sigma S(A)$$

Generative Molecules Can “Game” Scoring Functions



Dyachenko, Natalia V., et al. "Synthesis of fused heterocyclic systems via the Mallory photoreaction of arylthienylethenes." *Photochemical & Photobiological Sciences* 18.12 (2019): 2901-2911.



Article

Design of SARS-CoV-2 Main Protease Inhibitors Using Artificial Intelligence and Molecular Dynamic Simulations

Lars Elend ¹, Luise Jacobsen ², Tim Cofala ¹, Jonas Prellberg ¹, Thomas Teusch ³, Oliver Kramer ^{1,*} and Ilia A. Solov'yov ^{3,4,5,*}

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² Department of Physics, Chemistry and Pharmacy, University of Southern Denmark, Campusvej 55, 5230 Odense M, Denmark; luja@sdu.dk

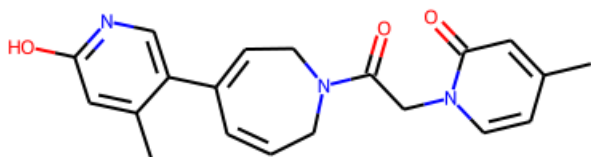
³ Department of Physics, Carl von Ossietzky University, Carl-von-Ossietzky-Str. 9-11, 26129 Oldenburg, Germany; thomas.teusch@uni-oldenburg.de

⁴ Research Center for Neurosensory Science, Carl von Ossietzky Universität Oldenburg, 26111 Oldenburg, Germany

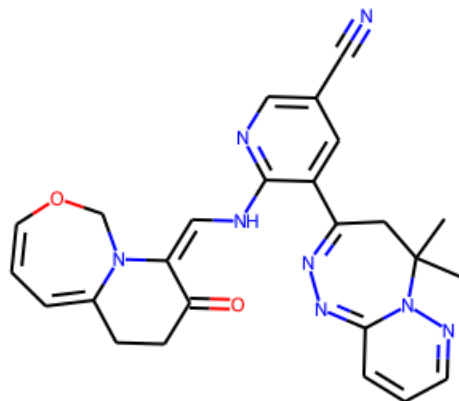
⁵ Center for Nanoscale Dynamics (CENAD), Carl von Ossietzky Universität Oldenburg, Institut für Physik, Ammerländer Heerstr. 114-118, 26129 Oldenburg, Germany

* Correspondence: oliver.kramer@uol.de (O.K.); ilia.solovyov@uni-oldenburg.de (I.A.S.); Tel.: +49-441-798-3817 (I.A.S.)

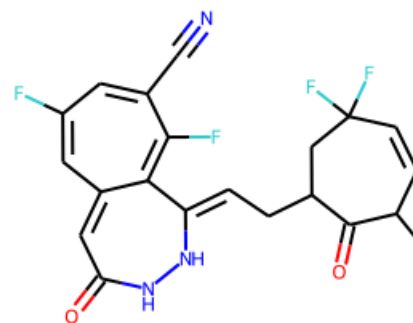
Highest Scoring Molecules Contain Questionable Functionality



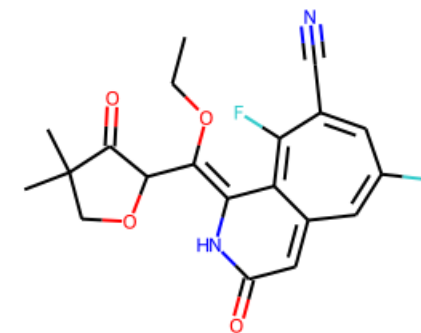
Lig 1



Lig 7

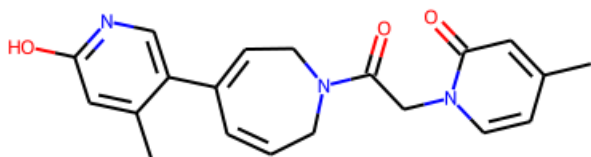


Lig 19

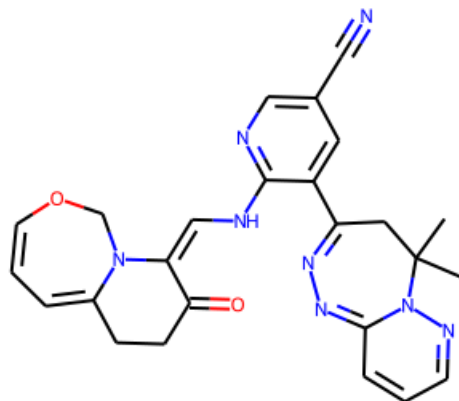


Lig 21

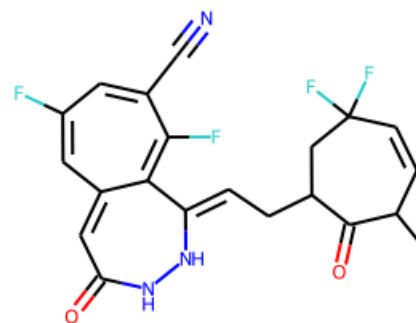
Highest Scoring Molecules Contain Questionable Functionality



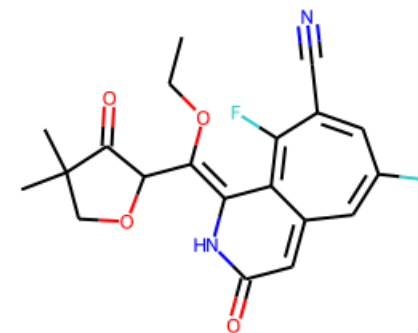
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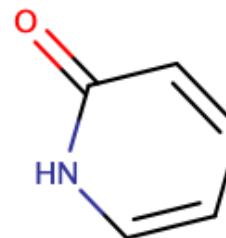
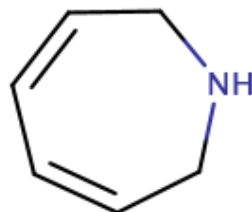
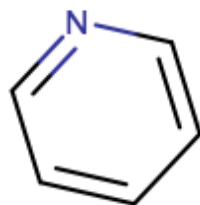
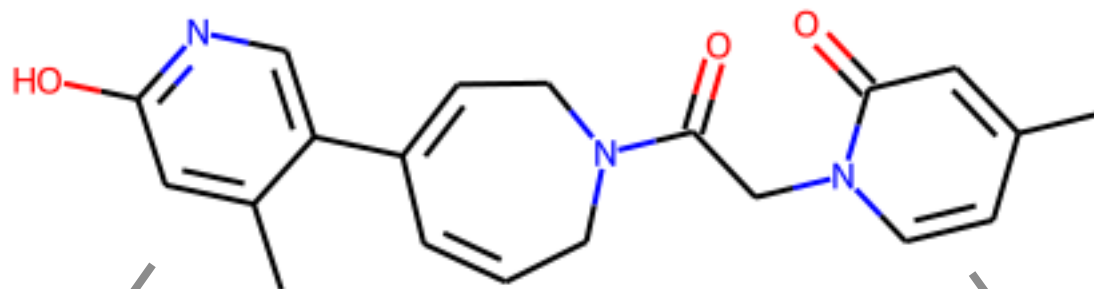


Lig 19



Lig 21

Evaluating Ring System Frequency



Ring

Occurrences in ChEMBL

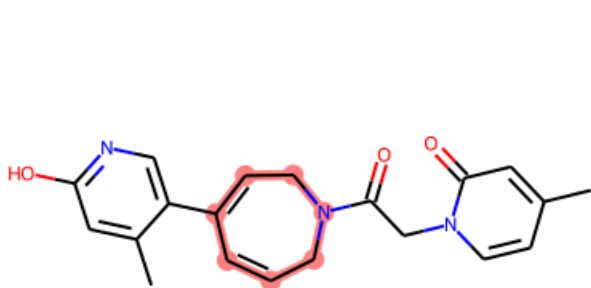
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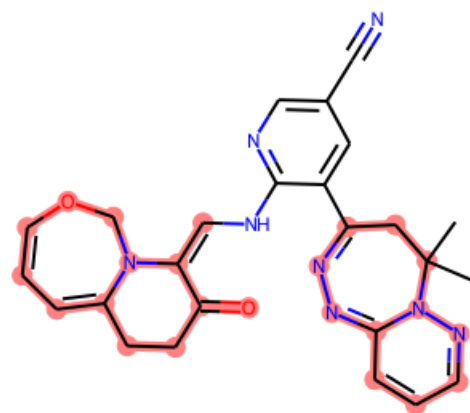
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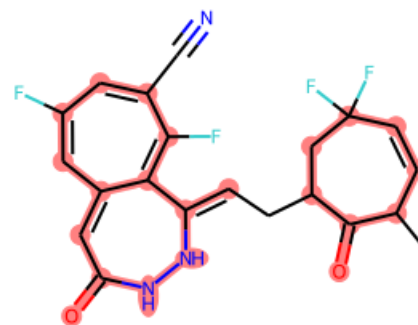
Top Molecules



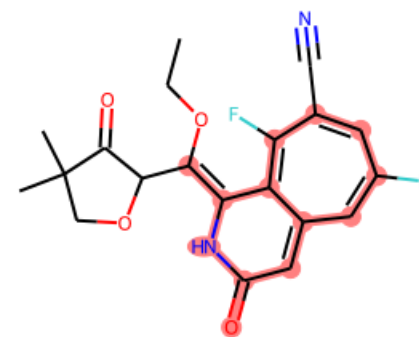
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Lig 7



Lig 19



Lig 21

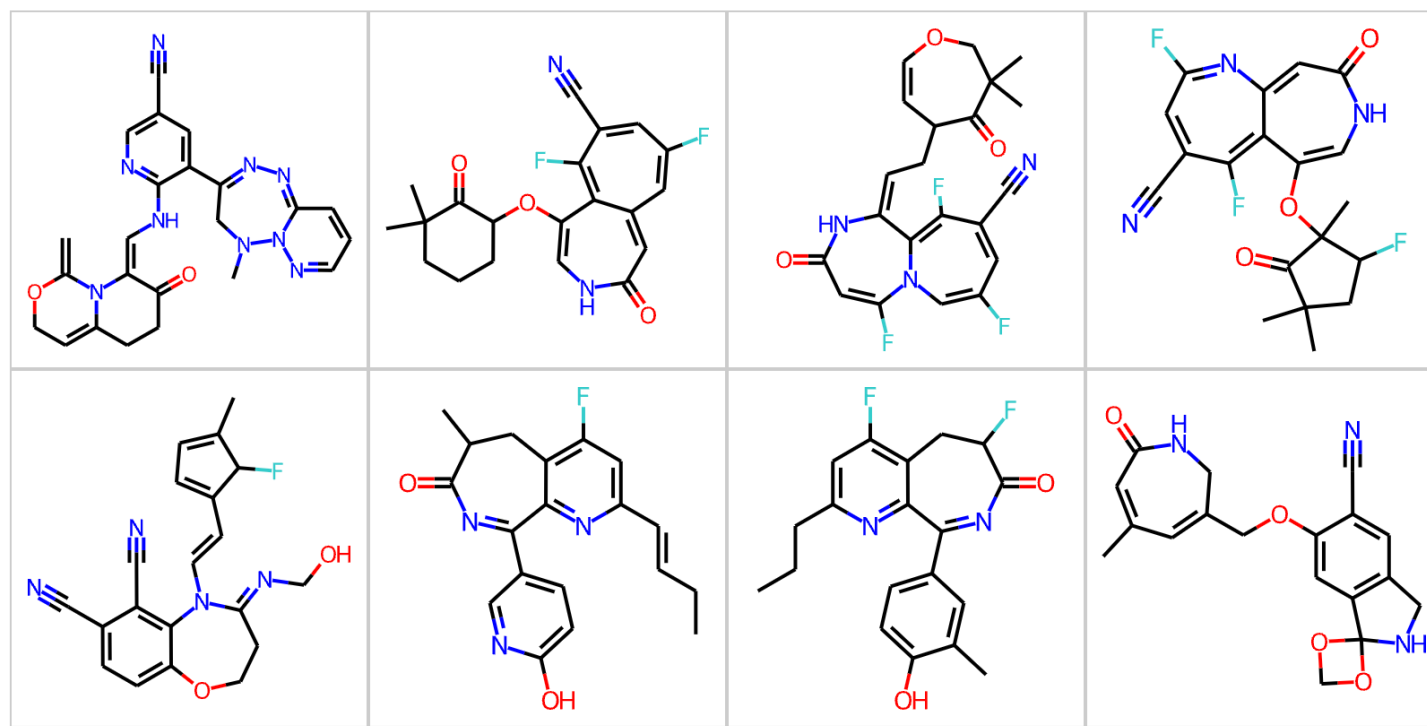
Evaluating Generative Model Output

144,350 molecules generated

23 molecules violated rules of valence

107,386 (74%) contained ring systems not found in ChEMBL

144,350 (79%) contained ring systems occurring < 10 times in ChEMBL



Article

Design of SARS-CoV-2 Main Protease Inhibitors Using Artificial Intelligence and Molecular Dynamic Simulations

Lars Elend ¹, Luise Jacobsen ², Tim Cofala ¹, Jonas Prellberg ¹, Thomas Teusch ³, Oliver Kramer ^{1,*} and Ilia A. Solov'yov ^{3,4,5,*}

¹ Computational Intelligence Lab, Department of Computer Science, Carl von Ossietzky University, Ammerländer Heerstraße 114-118, 26129 Oldenburg, Germany; lars.elend@uni-oldenburg.de (L.E.); tim.cofala@uni-oldenburg.de (T.C.); jonas.prellberg@uni-oldenburg.de (J.P.)

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³ Department of Physics, Carl von Ossietzky University, Carl-von-Ossietzky-Str. 9-11, 26129 Oldenburg, Germany; thomas.teusch@uni-oldenburg.de

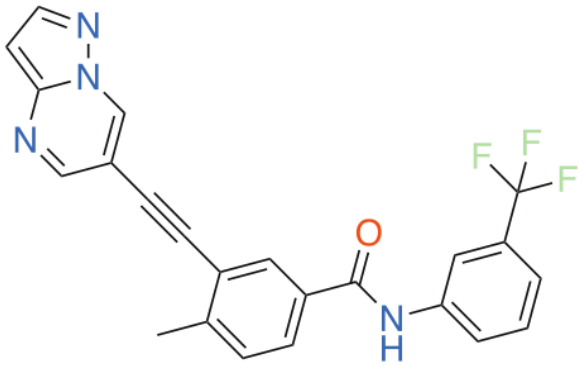
⁴ Research Center for Neurosensory Science, Carl von Ossietzky Universität Oldenburg, 26111 Oldenburg, Germany

⁵ Center for Nanoscale Dynamics (CENAD), Carl von Ossietzky Universität Oldenburg, Institut für Physik, Ammerländer Heerstr. 114-118, 26129 Oldenburg, Germany

* Correspondence: oliver.kramer@uol.de (O.K.); ilia.solov'yov@uni-oldenburg.de (I.A.S.); Tel.: +49-441-798-3817 (I.A.S.)

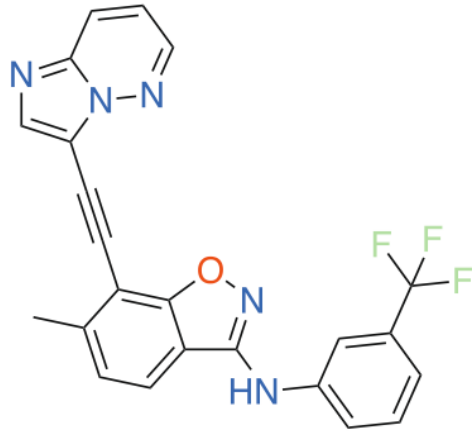
Molecules 2022, 27, 4020

Assessing the Novelty of Molecules Produced By Generative Models



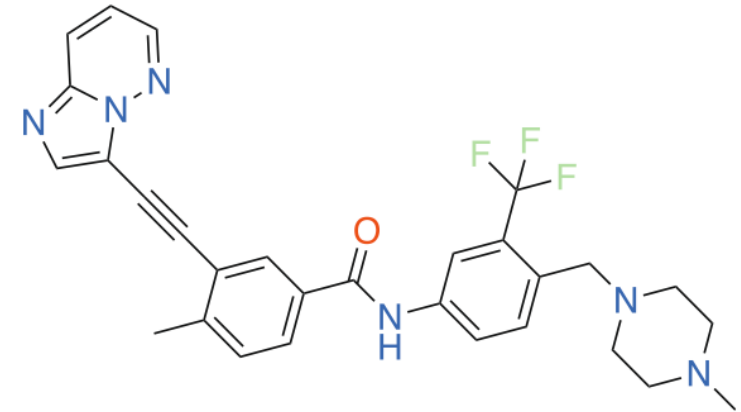
Gao et al.
Compound 7r
6 nM

Published in JMC 2013



Zhavoronkov et al.
Compound 1
10 nM

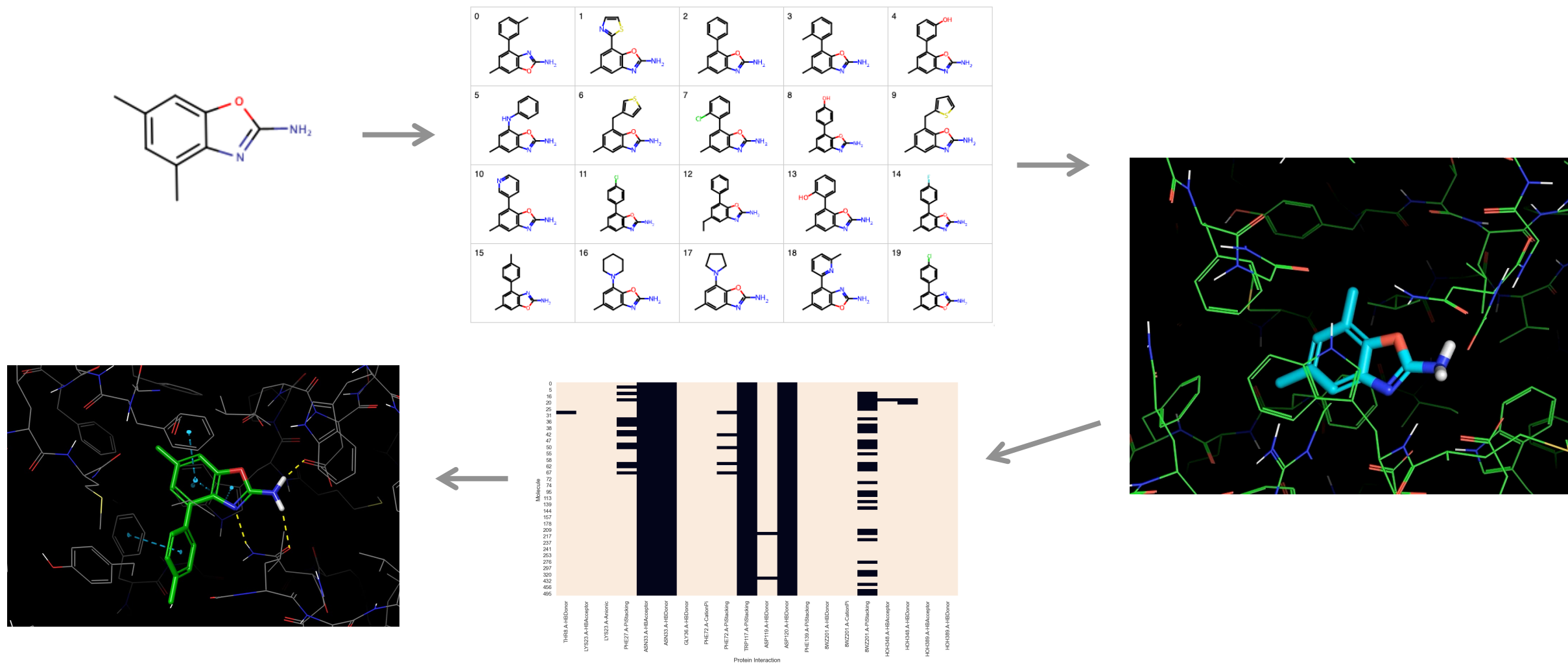
Generative Design
2020



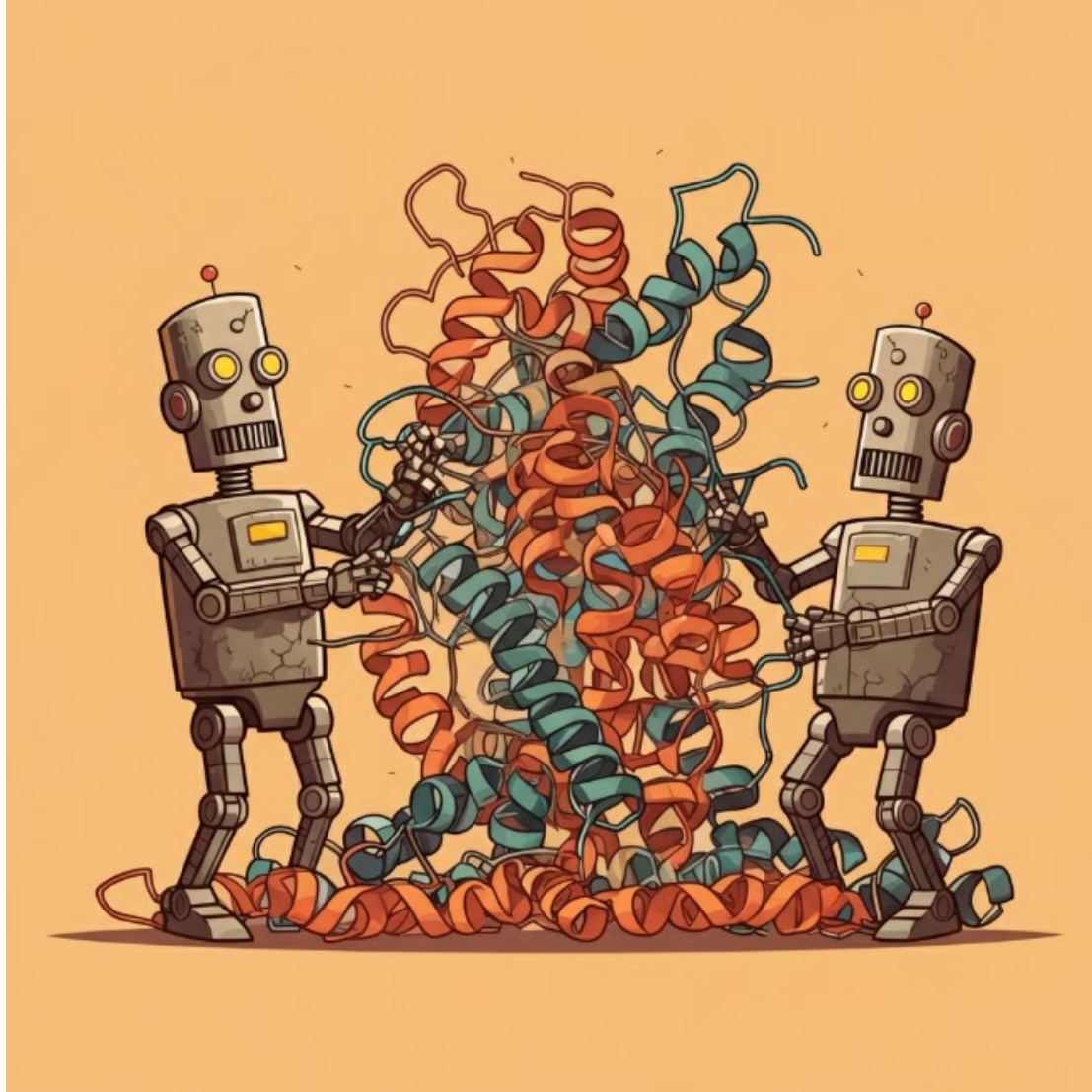
Ponatinib
9 nM

FDA Approved
2012

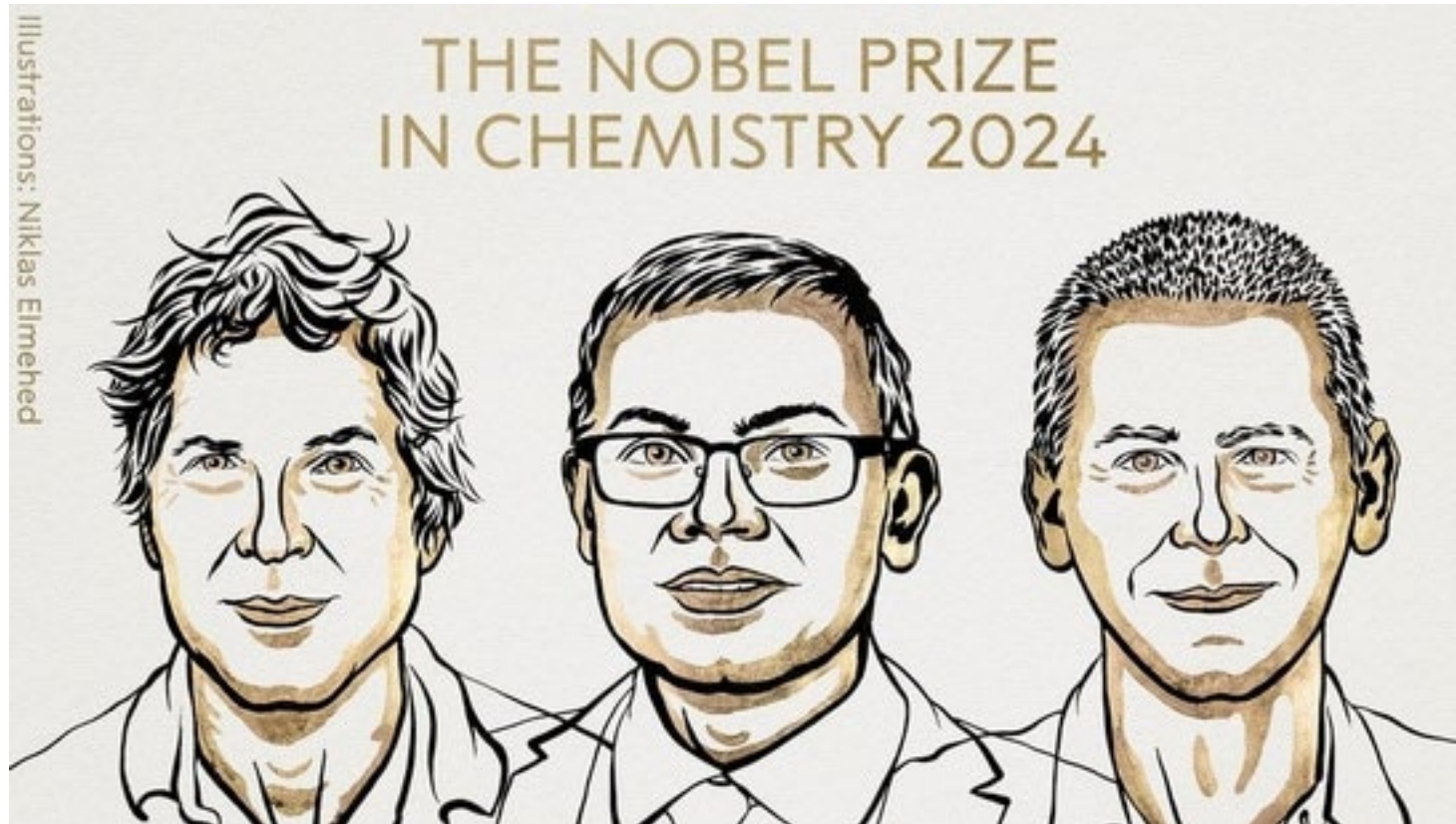
A Generative Design Workflow for Fragment Screening



Protein Structure Prediction and Cofolding

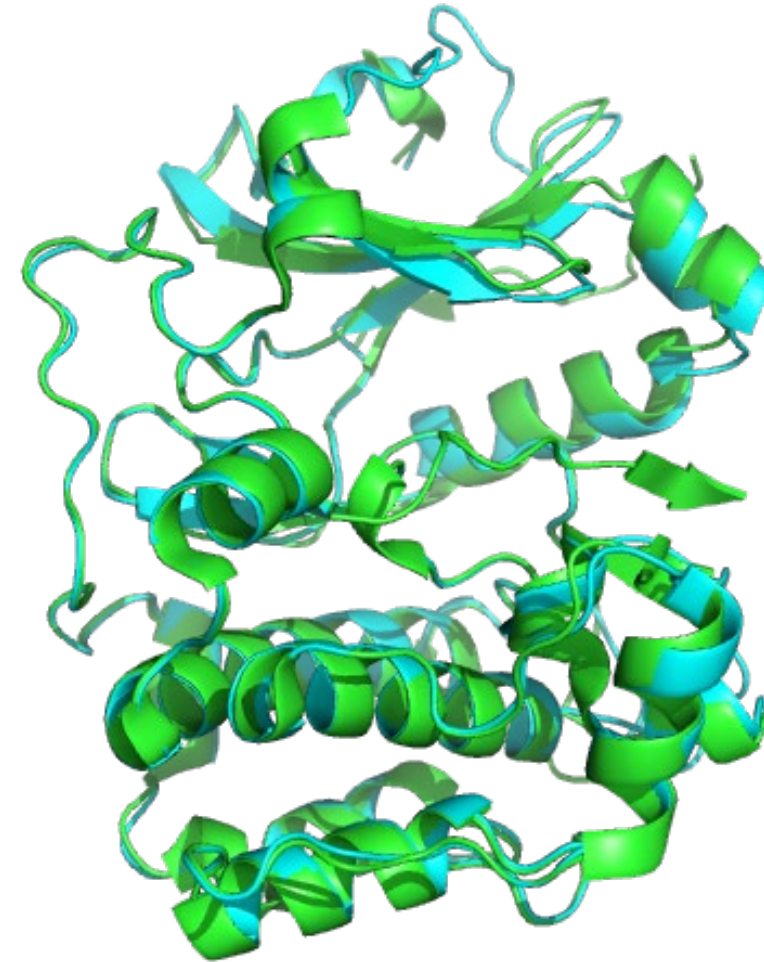


Protein Structure Prediction



Converting Amino Acid Sequence to 3D Structure

VTMNEFEYLKLLGKGTFGKVILVKEKATGRYYAMKI
LKKEVIVAKDEVAHTLTENRVLQNSRHPFLTALKYS
FQTHDRLCFVMEYANGGELFFHLSRERVFSEDRA
FYGAEIVSALDYLHSEKNVVYRDLKLENLMLDKDG
HIKITDFGLCKEGIKDGATMKFCGTPEYLAPEVLED
NDYGRAVDWWGLGVVMYEMMCGRLPFYNQD
HEKLFELILMEEIRFPRTLGPPEAKSLLSGLLKDPKQ
RLGGGSEDAKEIMQHRFFAGIVWQHVEKKLSPP
FKPQVTSETDTRYFDEEFTAQRPHFPQFDYSA

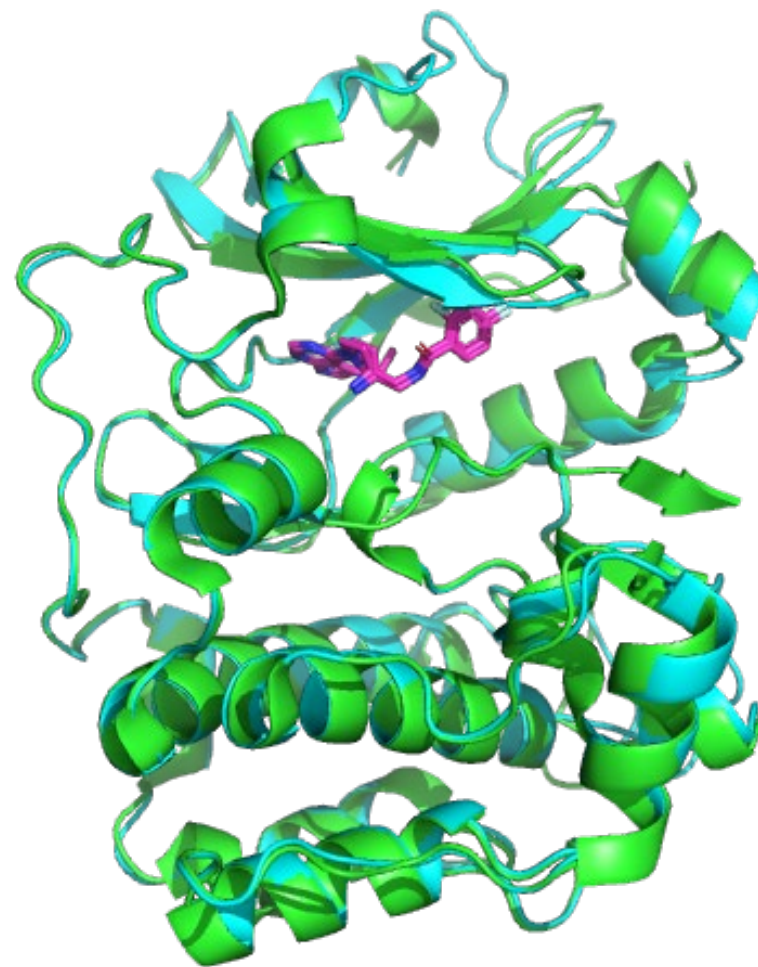


AlphaFold and related techniques convert sequence to 3D structure

Co-Folding Enables the Prediction of Ligand Poses With Protein Structures

VTMNEFEYLKLLGKGTFGKVILVKEKATGRYYAMKI
LKKEVIVAKDEVAHTLTENRVLQNSRHPFLTALKYS
FQTHDRLCFVMEYANGGELFFHLSRERVFSEDRA
FYGAEIVSALDYLHSEKNVVYRDLKLENLMLDKDG
HIKITDFGLCKEGIKDGATMKFCGTPEYLAPEVLED
NDYGRAVDWWGLGVVMYEMMCGRLPFYNQD
HEKLFELILMEEIRFPRTLGPPEAKSLLSGLLKDPKQ
RLGGGSEDAKEIMQHRFFAGIVWQHVEKKLSPP
FKPQVTSETDTRYFDEEFTAQRPHFPQFDYSA

CCc1c[nH]c2c1c(ncn2)N3CCC(C3)(CNC(=O)c4ccc(cc4F)F)N



Co-folding was introduced with DragonFold, AlphaFold 3, and RoseTTaFold All Atom

The Most Popular Slide at the 2025 CADD GRC

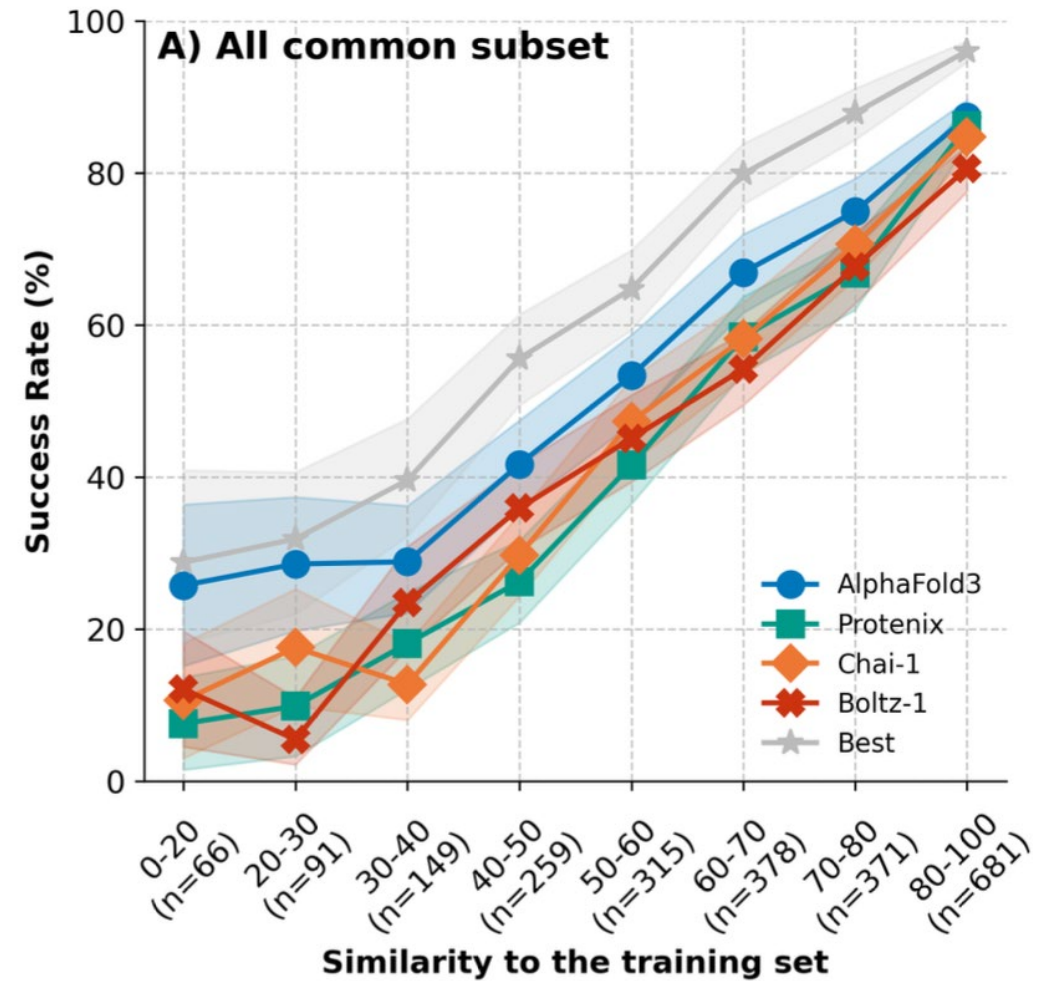
HAVE PROTEIN-LIGAND CO-FOLDING METHODS MOVED BEYOND MEMORISATION?

Peter Škrinjar
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Co-folding success rate correlates with training set similarity

Benchmarking Co-folding at Orthosteric and Allosteric Sites

Article

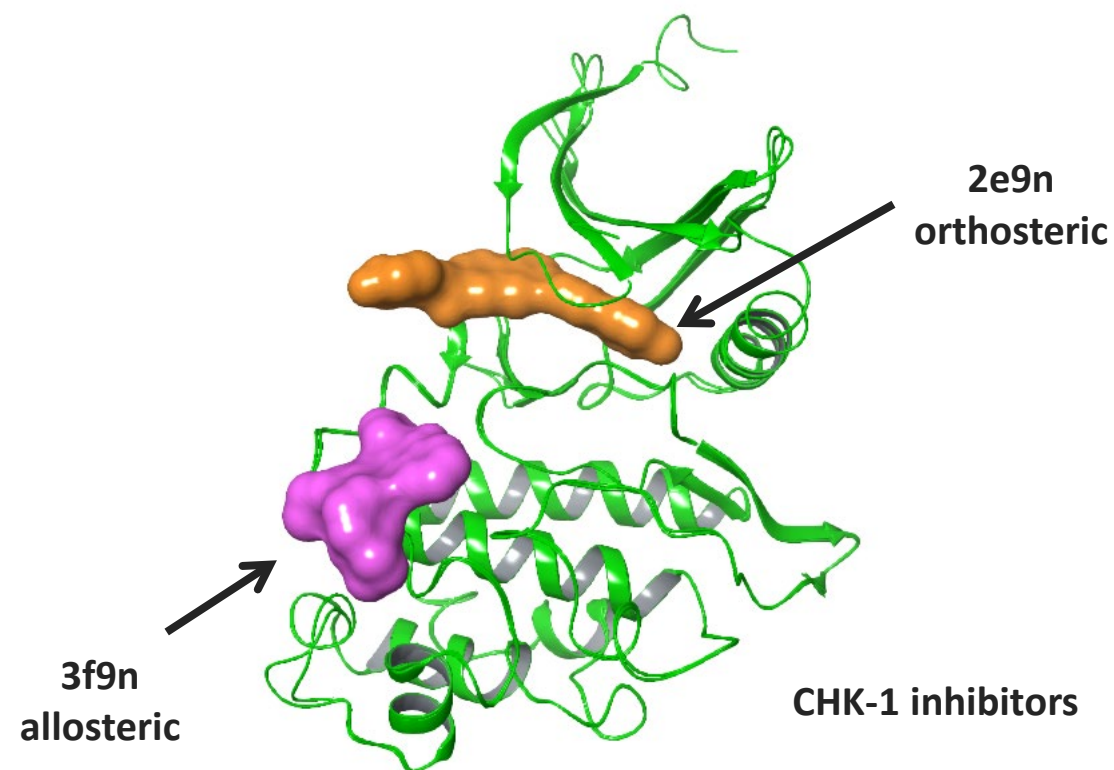
Co-folding, the Future of Docking – Prediction of Allosteric and Orthosteric Ligands

Eva Nittinger, Özge Yoluk, Alessandro Tibo, Gustav Olanders, Christian Tyrchan

This is a preprint; it has not been peer reviewed by a journal.

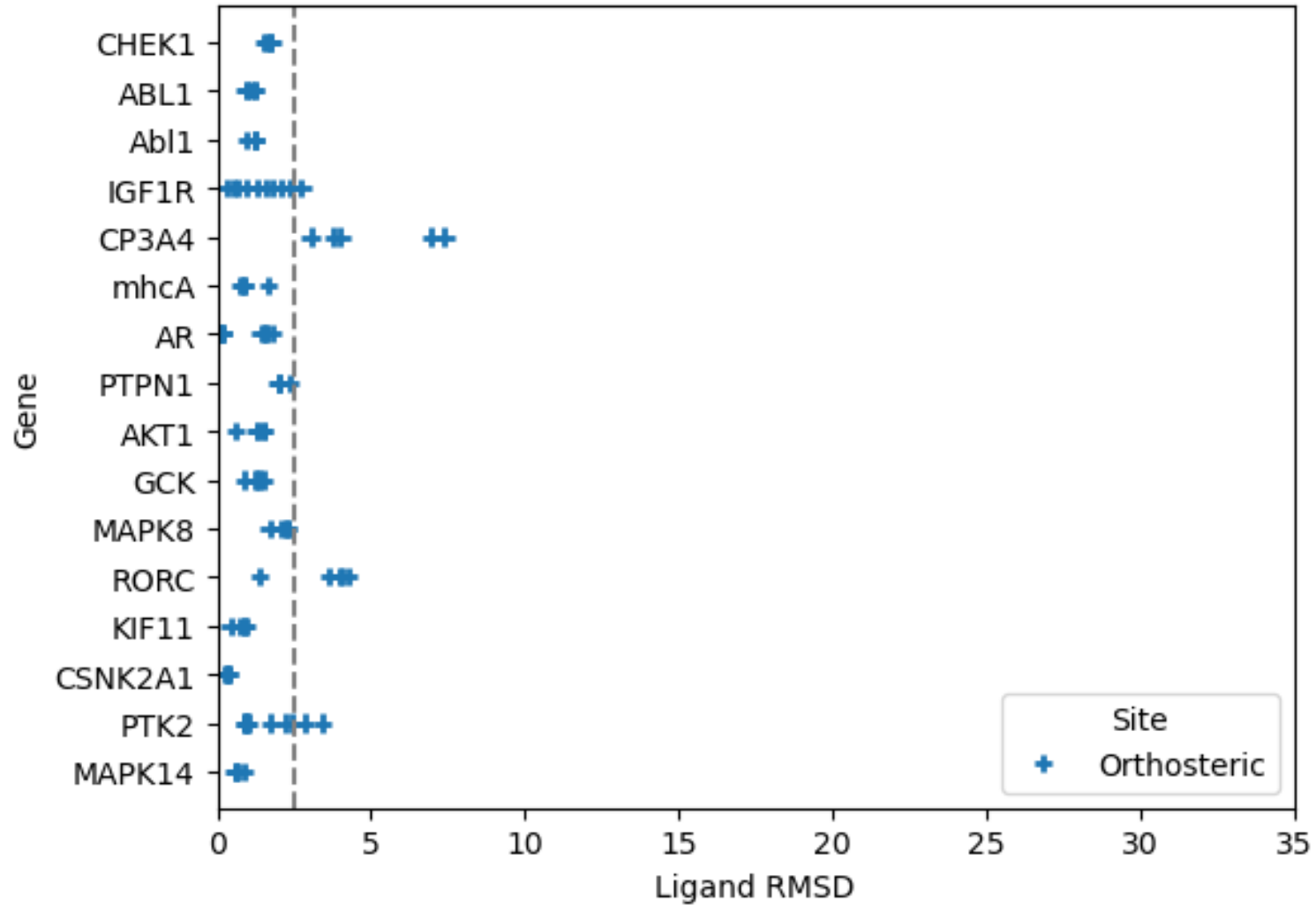
<https://doi.org/10.21203/rs.3.rs-6526650/v1>

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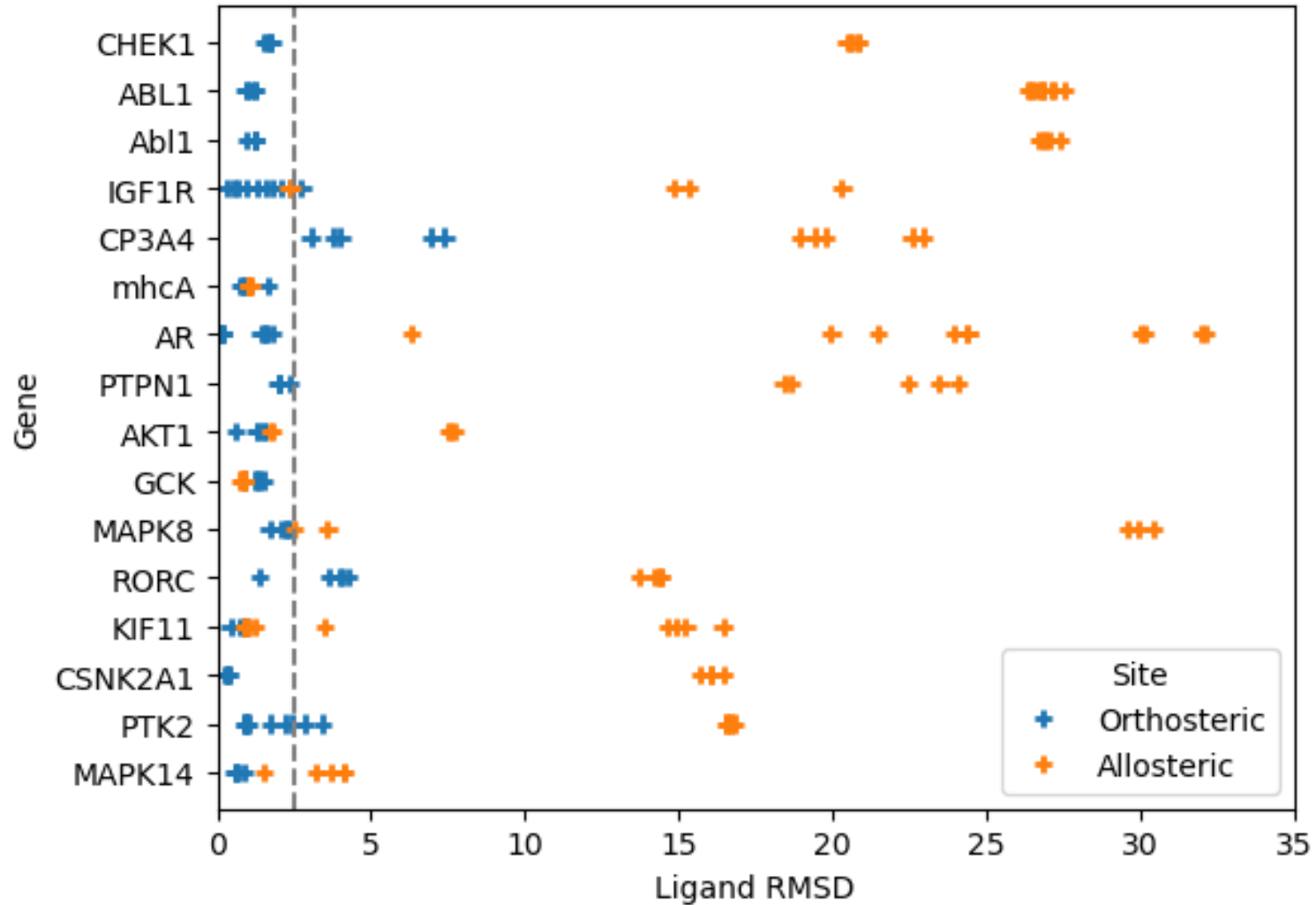
16 pairs protein/ligand complex with both allosteric and orthosteric binders

Co-Folding Performs Well With Orthosteric Binding Sites



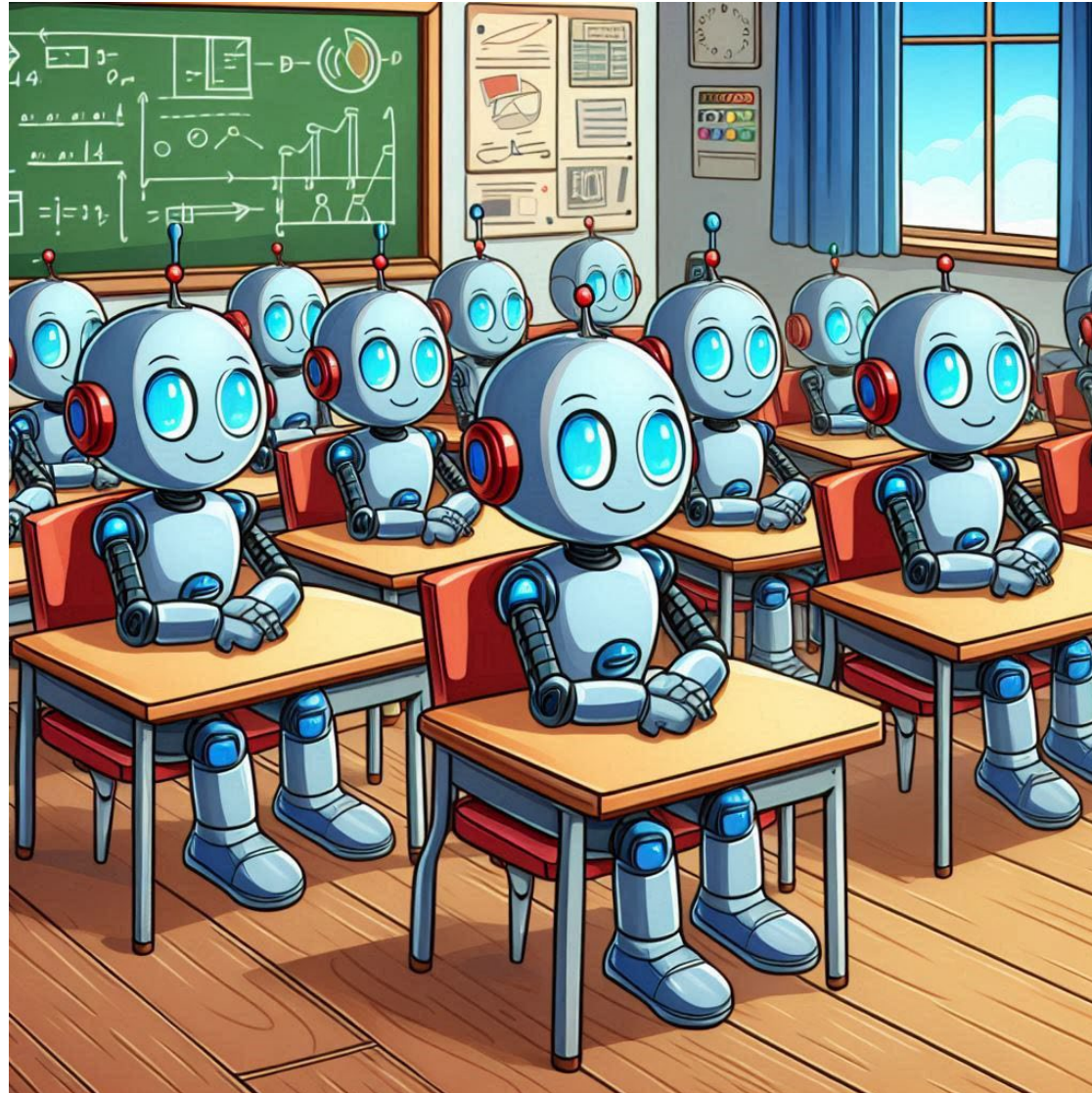
Study based on <https://www.researchsquare.com/article/rs-6526650/v1>

Co-Folding Performs Poorly With Allosteric Binding Sites

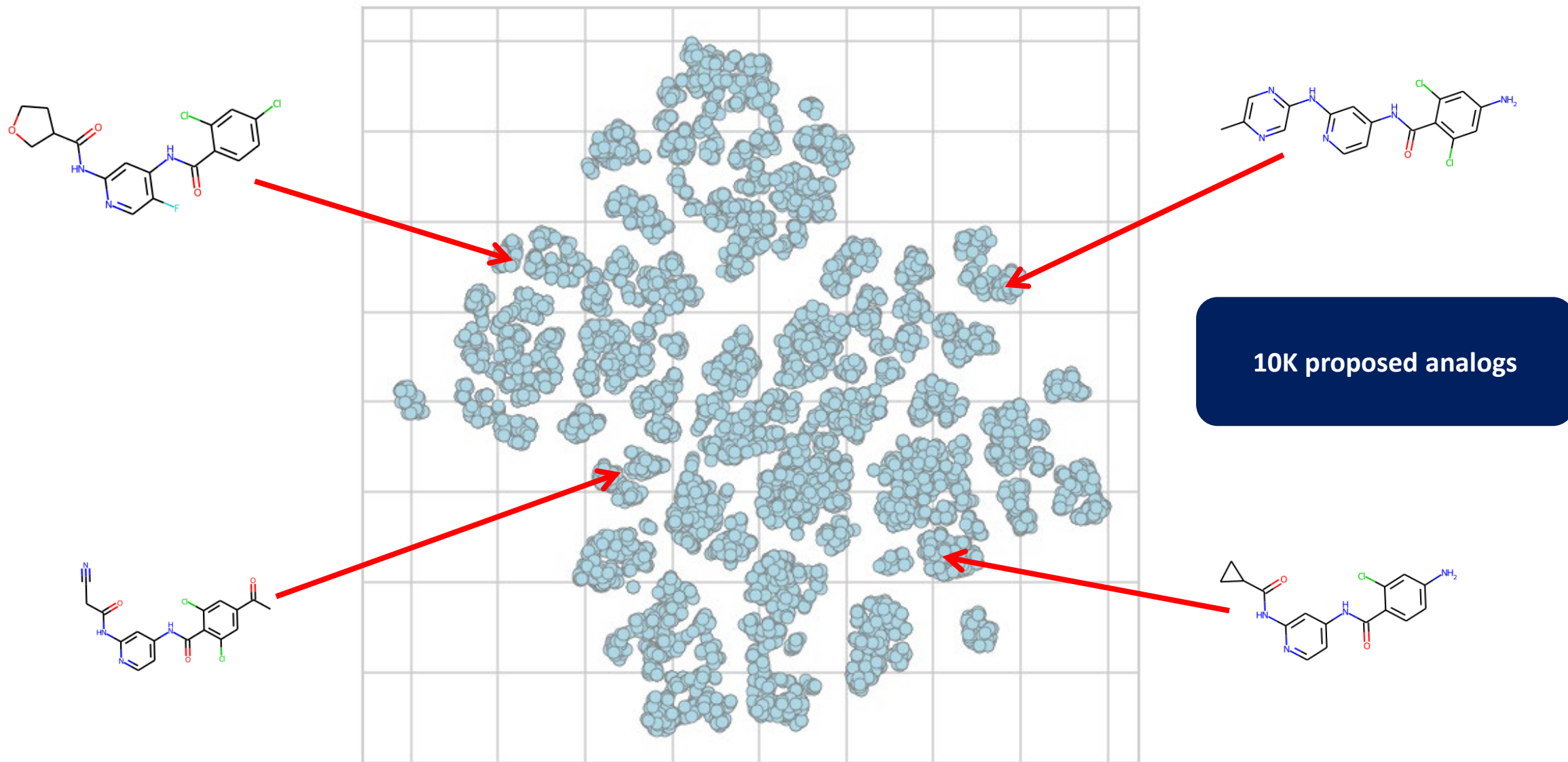


Study based on <https://www.researchsquare.com/article/rs-6526650/v1>

Active Learning



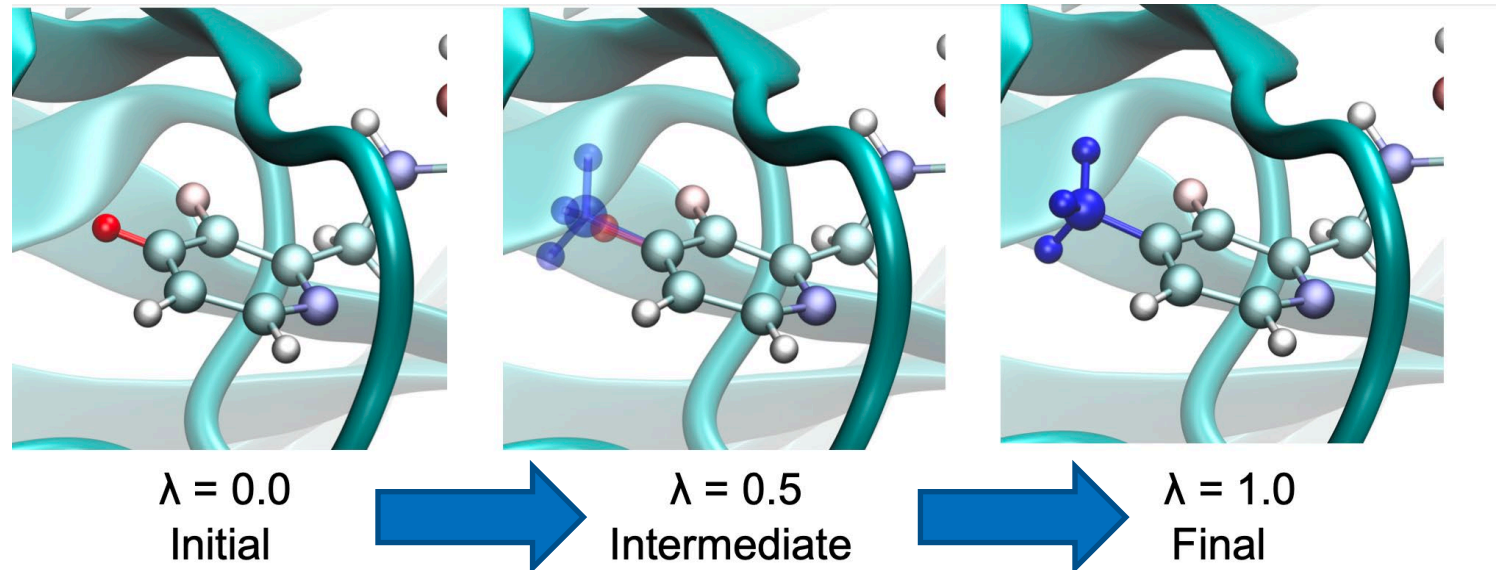
Prioritizing Molecules For Synthesis



Free Energy Perturbation (FEP): Transformation

FEP evaluates binding energy by morphing a known molecule into a new molecule

1. Alchemically morph one molecule to another
2. Run simulations using Molecular Dynamics
3. Compute free energy of transforming the molecule



Calculations typically take 4-8 hrs/molecule
10K calculations would require ~9 GPU years

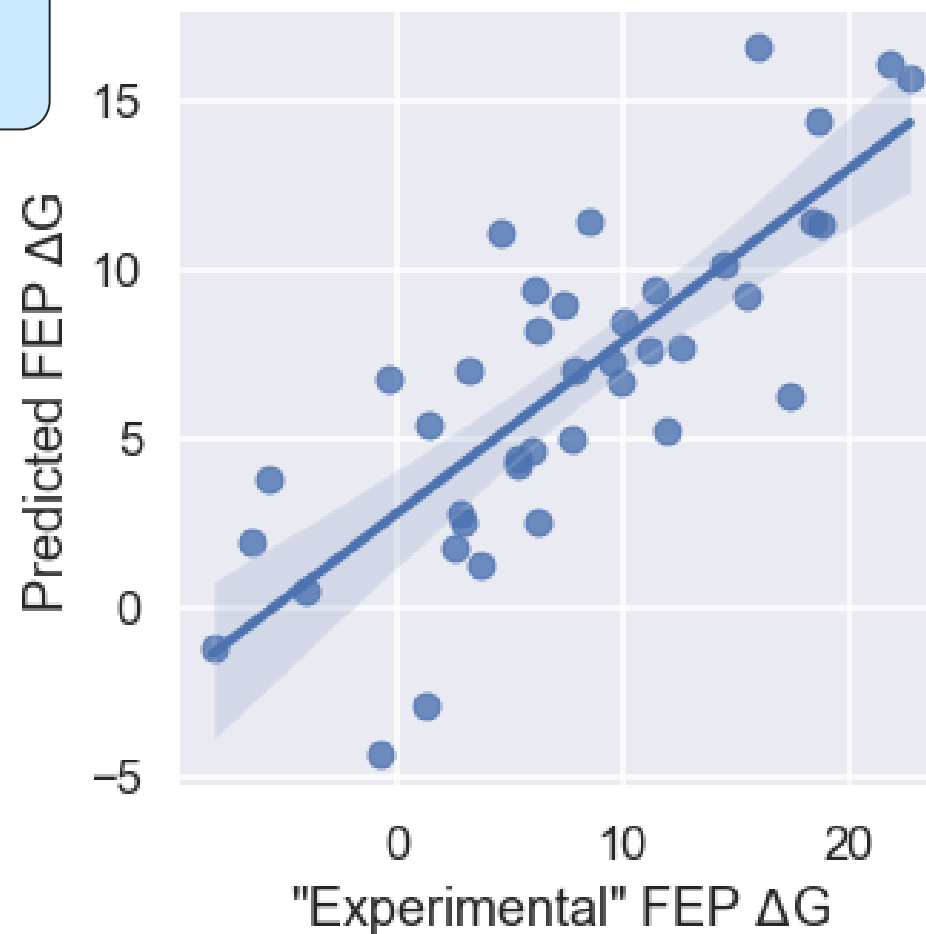
Build a Machine Learning Model to Predict FEP ΔG

Time to process 10K molecules

Method	Time (sec)
FEP	2.9×10^8
Machine Learning Model	7.5

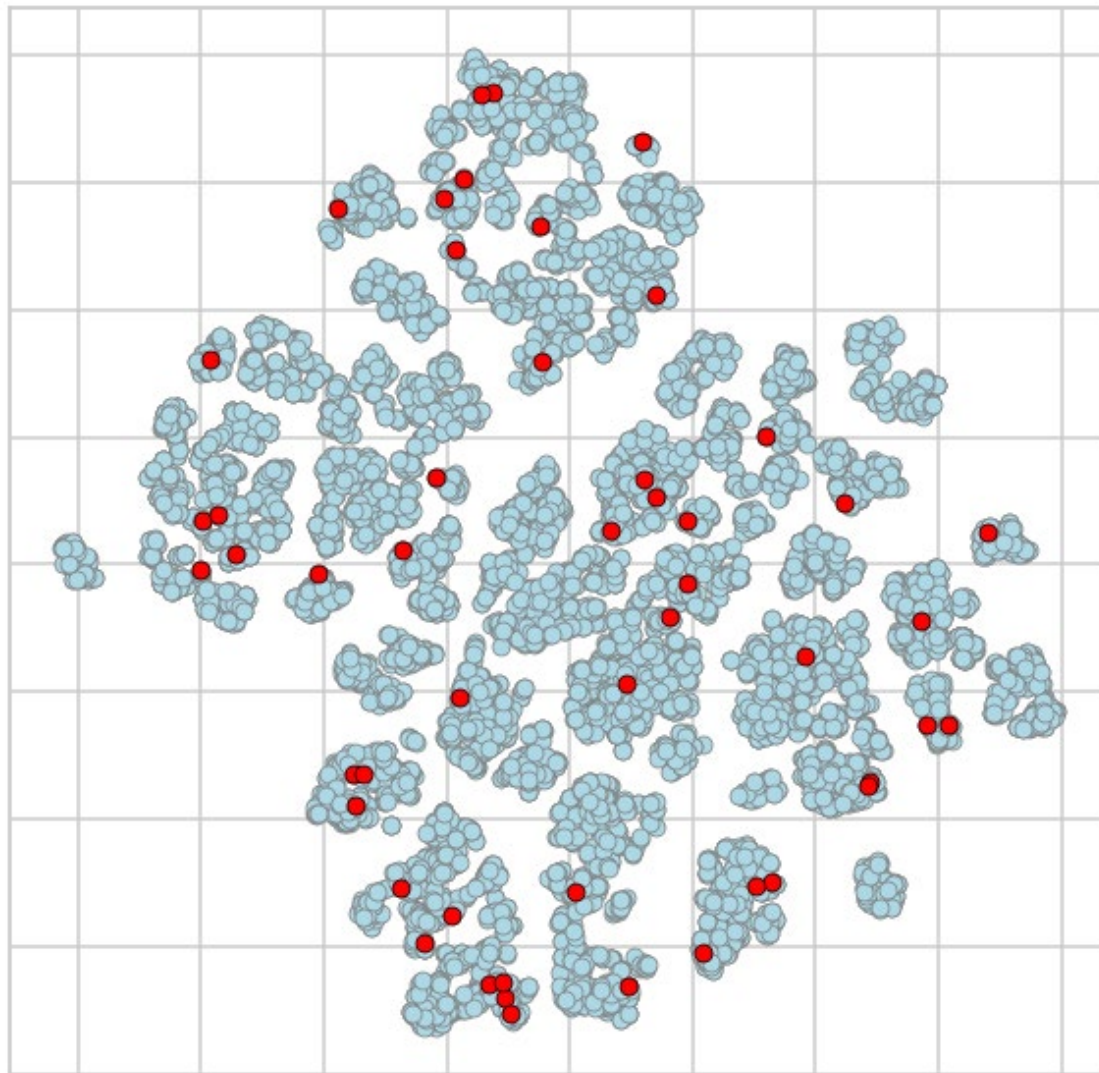
9 GPU years

Machine learning model is 39 million times faster than FEP*



*Training on 100 molecules + inference on 10K molecules

Sample 50 of 10K molecules and Run FEP



FEP



	SMILES	dG bind
112	<chem>COc1c(F)cccc1C(=O)Nc1cc(NC(=O)C2COC2)ncc1F</chem>	-12.83
275	<chem>CN(C)c1nccc(Nc2cc(NC(=O)c3c(F)ccc(F)c3Cl)ccn2)n1</chem>	-10.16
563	<chem>CCNC(=O)Nc1cc(NC(=O)c2cccc(O)c2Cl)ccn1</chem>	-7.68
589	<chem>O=C(CC1CC1)Nc1cc(NC(=O)c2cc(O)ccc2Cl)c(F)cn1</chem>	-7.51
1170	<chem>COC(=O)c1cccc(Nc2cc(NC(=O)c3cccc3Cl)c(F)cn2)n1</chem>	-4.58
1342	<chem>CNC(=O)Nc1cc(NC(=O)c2c(F)cccc2OC)ccn1</chem>	-3.84
1657	<chem>CC(C)NC(=O)Nc1cc(NC(=O)c2c(F)ccc(F)c2Cl)ccn1</chem>	-2.71
1664	<chem>Cc1cccc(Cl)c1C(=O)Nc1cc(Nc2ccnnc2)ncc1F</chem>	-2.69
1974	<chem>O=C(Nc1cc(NC(=O)C2CCCO2)ncc1F)c1c(F)cccc1Br</chem>	-1.80
2071	<chem>O=C(Nc1cc(NC(=O)c2c(Cl)cccc2Br)ccn1)NC1CC1</chem>	-1.51
2372	<chem>Cc1cc(Cl)c(C(=O)Nc2ccnc(Nc3cc(C)nc(CO)n3)c2)c(...</chem>	-0.74
2542	<chem>Cc1cc(Nc2cc(NC(=O)c3cccc3C)ccn2)nc(NC2CC2)n1</chem>	-0.32
2792	<chem>CSCC(=O)Nc1cc(NC(=O)c2c(Cl)cc(N)cc2Cl)c(F)cn1</chem>	0.31
2815	<chem>COC(=O)Nc1cc(NC(=O)c2cccc(F)c2Cl)c(F)cn1</chem>	0.37
2880	<chem>COc1ccc(C(=O)Nc2cc(NC(=O)C3CC3(F)F)ncc2F)c(Cl)c1</chem>	0.54
3265	<chem>COc1ccc(Cl)c(C(=O)Nc2ccnc(NC(=O)C3CC3CO)c2)c1</chem>	1.45
3355	<chem>O=C(Nc1ccnc(NC(=O)C2CC2(F)F)c1)c1ccc(O)cc1F</chem>	1.65
3568	<chem>CC1CC1C(=O)Nc1cc(NC(=O)c2c(F)cc(F)cc2F)ccn1</chem>	2.19
3829	<chem>O=C(Nc1cc(Nc2ccnnc2)ncc1F)c1cccc(Cl)c1</chem>	2.76
3857	<chem>COc1ccc(F)c(C(=O)Nc2cc(NC(=O)C3CC3C)ncc2F)c1</chem>	2.82
4644	<chem>COc1cccc(C(=O)Nc2cc(NC(=O)C3CC(=O)C3)ncc2F)c1F</chem>	4.45
4729	<chem>Cc1nc(Cl)cc(Nc2cc(NC(=O)c3ccc(N)cc3)ccn2)n1</chem>	4.65
4901	<chem>CN(C)C(=O)c1cccc(Nc2cc(NC(=O)c3cccc(N)c3Cl)ccn...</chem>	5.00
4987	<chem>COc1ccc(Nc2cc(NC(=O)c3c(Cl)cccc3Br)ccn2)n1</chem>	5.13

etc.

Generate Molecular Descriptors From Chemical Structures

SMILES		Molecular Descriptors																				
	SMILES	0	1	2	3	4	5	6	7	8	9	...	90	91	92	93	94	95	96	97	98	99
112	COc1c(F)cccc1C(=O)Nc1cc(NC(=O)C2COC2)ncc1F	2.266	-2.240	2.246	-2.381	6.062	-0.133	19.144	10.089	1.784	855.897	...	23.320	0.0	4.984	5.918	0.000	30.957	47.660	0.0	5.75	15.370
275	CN(C)c1nccc(Nc2cc(NC(=O)c3c(F)ccc(F)c3Cl)ccn2)n1	2.192	-2.121	2.244	-2.152	6.342	0.102	35.496	10.155	1.797	1030.708	...	40.780	0.0	14.952	0.000	0.000	29.629	64.946	0.0	0.00	15.533
563	CCNC(=O)Nc1cc(NC(=O)c2cccc(O)c2Cl)ccn1	2.133	-2.093	2.269	-2.281	6.354	0.102	35.496	10.165	2.122	736.412	...	35.044	0.0	10.301	0.000	6.924	17.178	47.115	0.0	5.75	15.950
589	O=C(CC1CC1)Nc1cc(NC(=O)c2cc(O)ccc2Cl)c(F)cn1	2.243	-2.100	2.322	-2.159	6.341	-0.116	35.496	10.163	1.742	839.883	...	34.921	0.0	4.984	5.918	19.262	10.634	46.866	0.0	5.75	10.634
1170	COC(=O)c1cccc(Nc2cc(NC(=O)c3ccccc3Cl)c(F)cn2)n1	2.139	-2.103	2.221	-2.138	6.340	0.059	35.496	10.177	1.811	1045.192	...	40.801	0.0	9.968	0.000	0.000	17.743	76.825	0.0	0.00	10.634
1342	CNC(=O)Nc1cc(NC(=O)c2c(F)cccc2OC)ccn1	2.175	-2.107	2.225	-2.246	6.062	0.102	19.142	10.142	2.210	736.412	...	23.444	0.0	10.301	0.000	0.000	24.791	47.909	0.0	5.75	20.687
1657	CC(C)NC(=O)Nc1cc(NC(=O)c2c(F)ccc(F)c2Cl)ccn1	2.193	-2.112	2.239	-2.339	6.342	0.102	35.496	10.155	2.159	814.717	...	35.044	0.0	10.301	0.000	19.889	10.634	52.683	0.0	0.00	15.950
1664	Cc1cccc(Cl)c1C(=O)Nc1cc(Nc2ccnccn2)ncc1F	2.155	-2.112	2.235	-2.151	6.342	0.102	35.496	10.087	1.847	899.917	...	34.831	0.0	14.952	0.000	6.924	10.634	71.020	0.0	0.00	10.634
1974	O=C(Nc1cc(NC(=O)C2CCCO2)ncc1F)c1c(F)cccc1Br	2.241	-2.121	2.233	-2.270	9.103	-0.124	79.919	10.164	1.771	836.892	...	39.250	0.0	4.984	0.000	18.946	17.240	52.133	0.0	0.00	10.634
2071	O=C(Nc1cc(NC(=O)c2c(Cl)cccc2Br)ccn1)NC1CC1	2.229	-2.101	2.257	-2.303	9.103	0.102	79.919	10.177	1.693	775.147	...	50.974	0.0	10.301	0.000	18.883	10.634	51.587	0.0	0.00	15.950
2372	Cc1cc(Cl)c(C(=O)Nc2ccnc(Nc3cc(C)nc(CO)n3)c2)c(...	2.158	-2.111	2.274	-2.134	6.401	0.102	35.497	10.124	1.804	1019.578	...	46.432	0.0	14.952	0.000	20.454	10.634	69.219	0.0	0.00	10.634
2542	Cc1cc(Nc2cc(NC(=O)c3ccccc3C)ccn2)nc(NC2CC2)n1	2.218	-2.108	2.243	-2.132	6.049	0.102	16.149	10.091	1.486	1015.283	...	29.179	0.0	14.952	0.000	32.731	15.950	65.482	0.0	0.00	15.950
2792	CSCC(=O)Nc1cc(NC(=O)c2c(Cl)cc(N)cc2Cl)c(F)cn1	2.178	-2.111	2.288	-2.171	7.988	-0.113	35.497	10.154	2.248	812.169	...	63.971	0.0	4.984	0.000	0.000	28.376	45.822	0.0	0.00	16.367
2815	COC(=O)Nc1cc(NC(=O)c2cccc(F)c2Cl)c(F)cn1	2.159	-2.095	2.225	-2.159	6.341	0.102	35.496	10.168	2.248	770.928	...	35.107	0.0	4.984	0.000	0.000	17.743	52.683	0.0	0.00	10.634
2880	COc1ccc(C(=O)Nc2cc(NC(=O)C3CC3(F)F)ncc2F)c(Cl)c1	2.609	-2.104	2.486	-2.189	6.342	-0.119	35.496	10.165	1.745	923.537	...	34.921	0.0	4.984	5.918	12.343	17.743	46.866	0.0	5.75	15.370
3265	COc1ccc(Cl)c(C(=O)Nc2ccnc(NC(=O)C3CC3CO)c2)c1	2.419	-2.098	2.382	-2.170	6.341	-0.117	35.496	10.163	1.685	842.936	...	34.921	0.0	4.984	11.836	6.421	24.350	47.115	0.0	5.75	15.370
3355	O=C(Nc1ccnc(NC(=O)C2CC2(F)F)c1)c1ccc(O)cc1F	2.609	-2.085	2.485	-2.185	6.045	-0.119	19.287	10.157	1.690	857.514	...	23.320	0.0	4.984	5.918	12.343	10.634	47.909	0.0	5.75	10.634
3568	CC1CC1C(=O)Nc1cc(NC(=O)c2c(F)cc(F)cc2F)ccn1	2.387	-2.090	2.379	-2.176	6.046	-0.117	19.149	10.146	1.737	833.777	...	23.320	0.0	4.984	11.836	13.345	10.634	53.477	0.0	0.00	10.634
3829	O=C(Nc1cc(Nc2ccnccn2)ncc1F)c1cccc(Cl)c1	2.113	-2.093	2.204	-2.132	6.306	0.102	35.496	10.179	1.800	875.352	...	34.831	0.0	14.952	0.000	0.000	10.634	71.523	0.0	0.00	10.634
3857	COc1ccc(F)c(C(=O)Nc2cc(NC(=O)C3CC3C)ncc2F)c1	2.388	-2.102	2.380	-2.175	6.048	-0.117	19.144	10.152	1.773	872.755	...	23.320	0.0	4.984	11.836	13.345	17.743	47.660	0.0	5.75	15.370

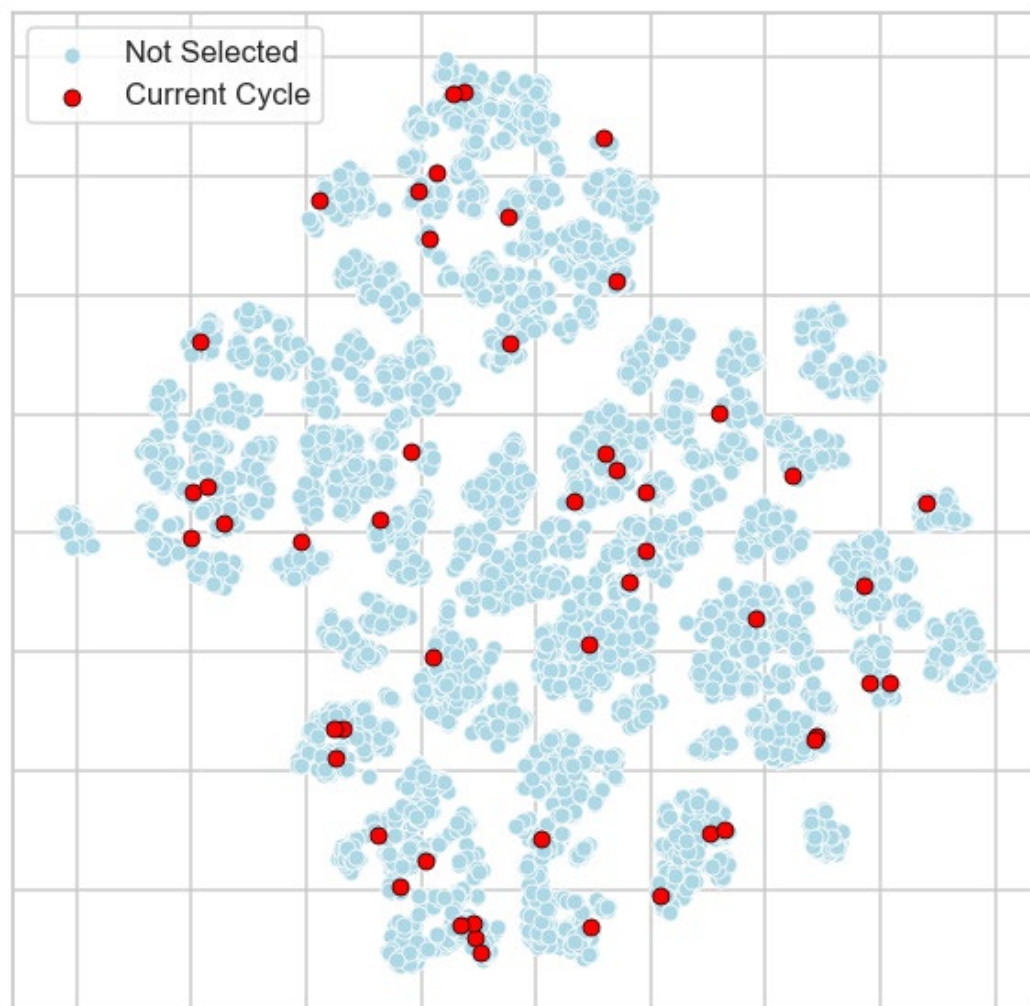
$$\Delta G = f(\text{molecular descriptors})$$

Predict ΔG for the remaining 9,950 molecules

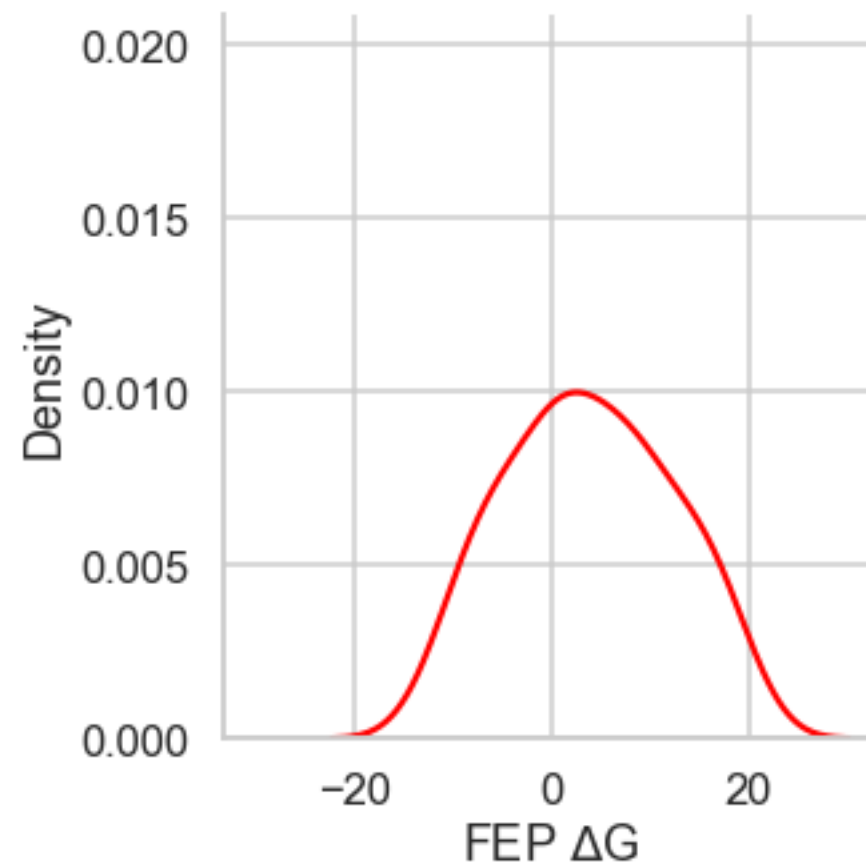
Run FEP on the 50 molecules with the best predicted ΔG

Repeat

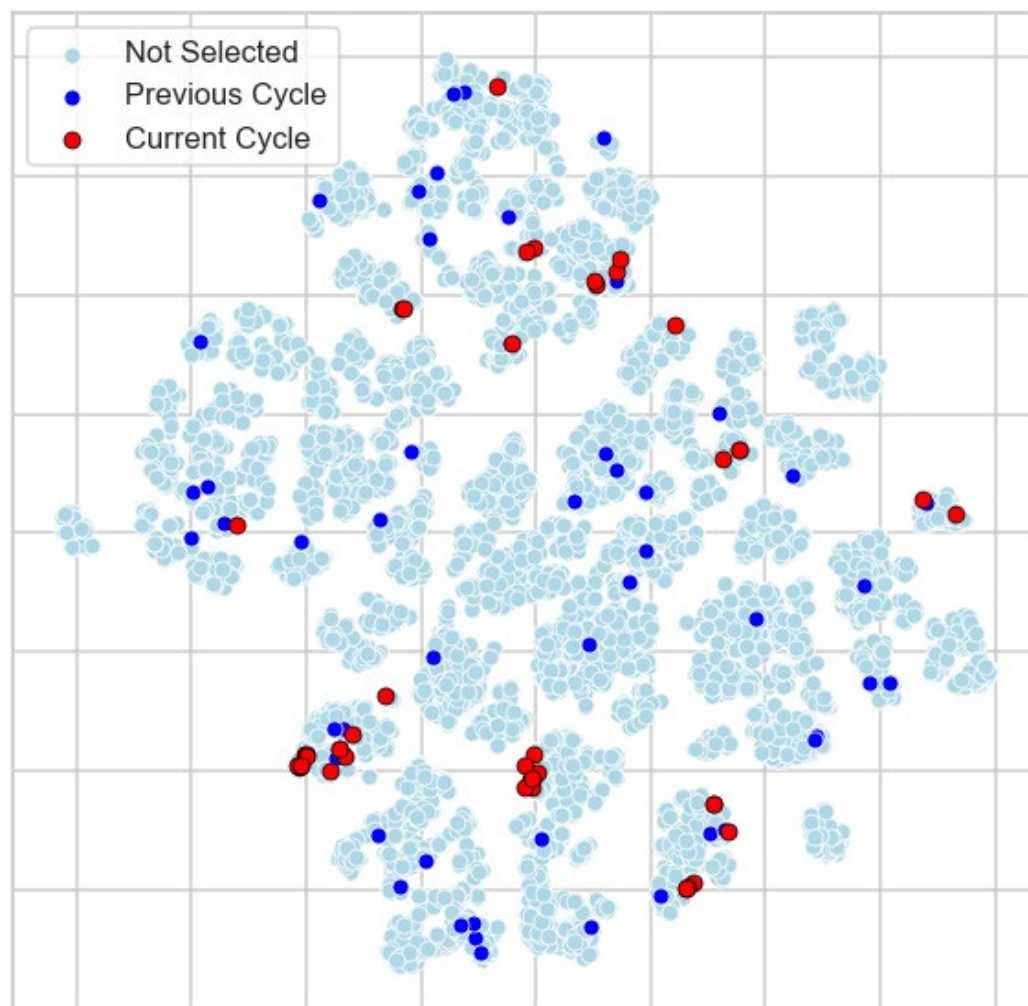
Initial Sample - 1 of the Top 100 Molecules Found



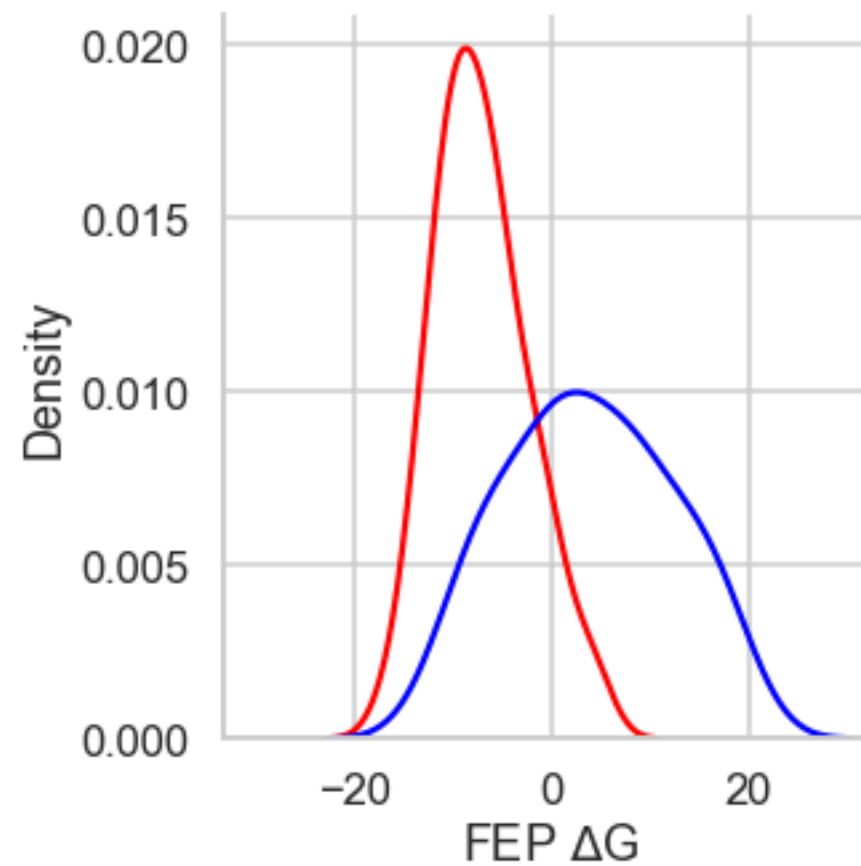
Sampled 0.5%



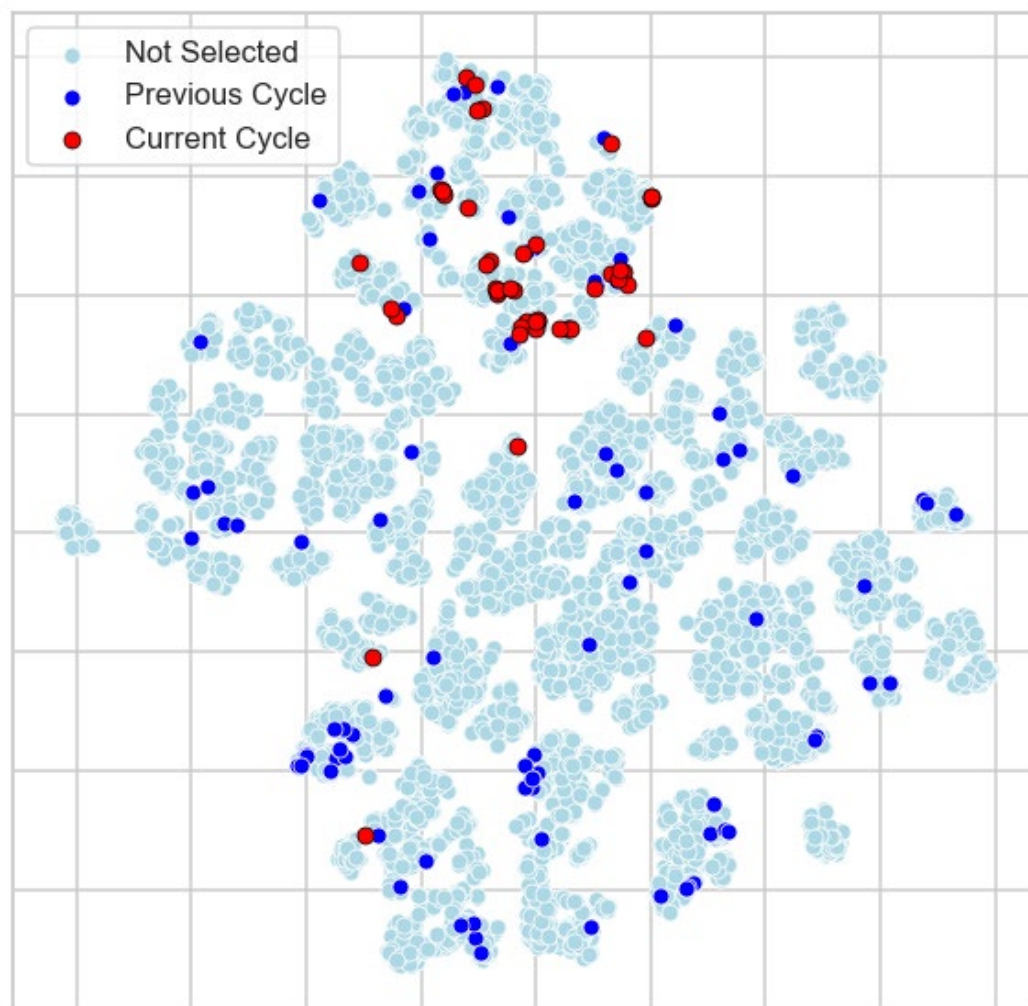
Cycle 1 - 16 of the Top 100 Molecules Found



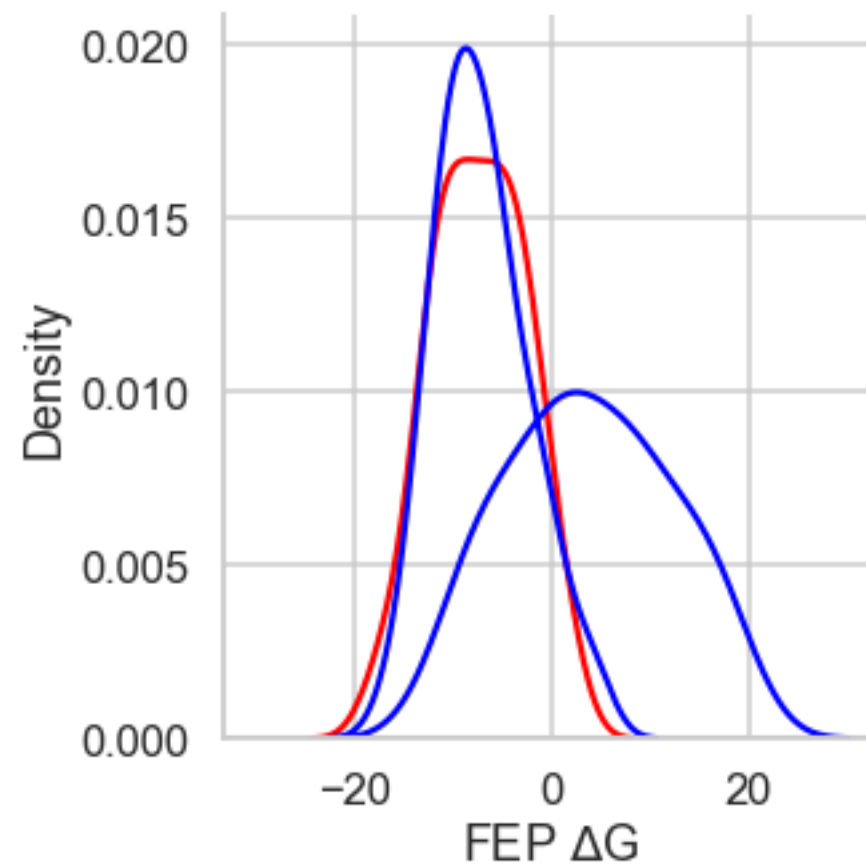
Sampled 1%



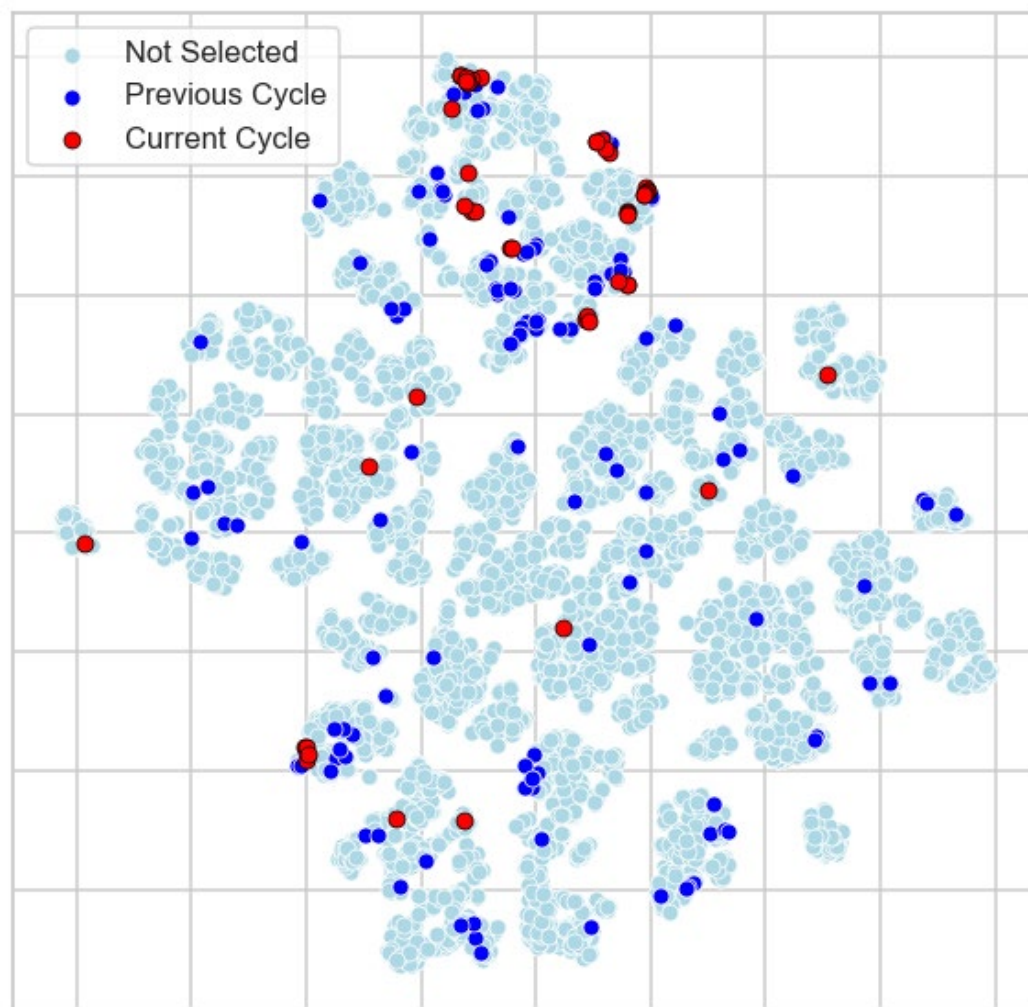
Cycle 2 - 31 of the Top 100 Molecules Found



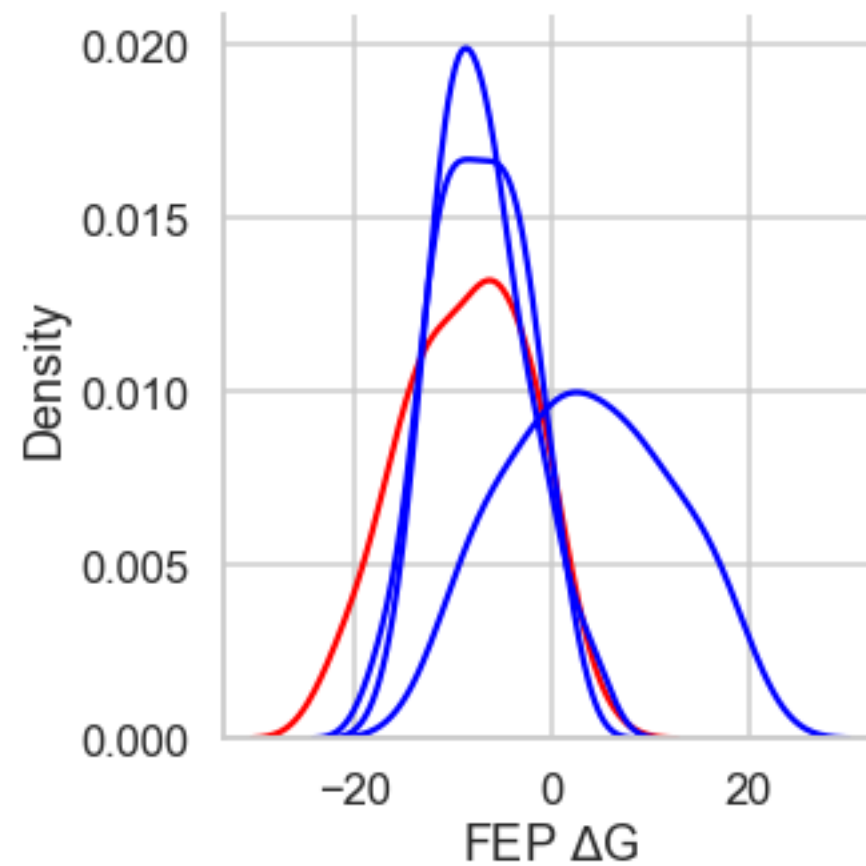
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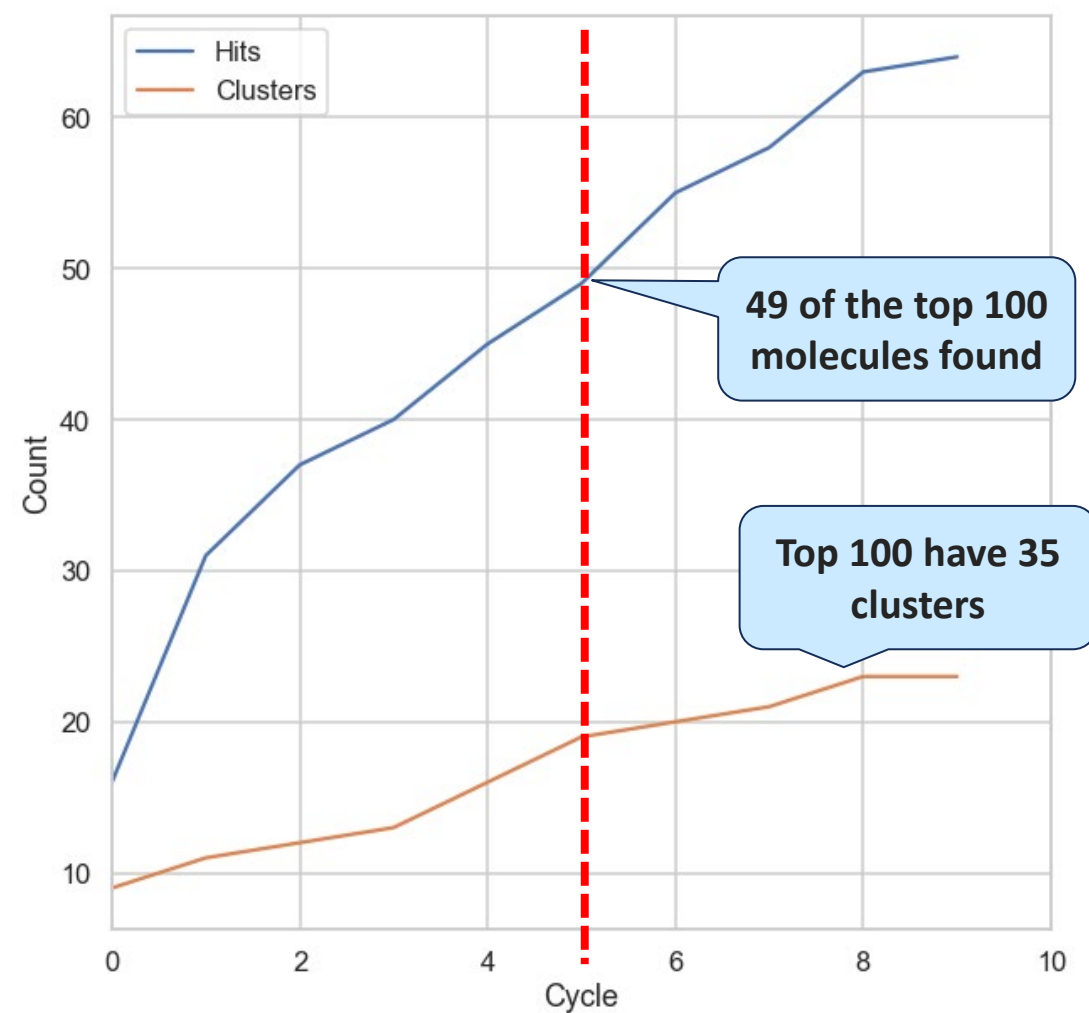
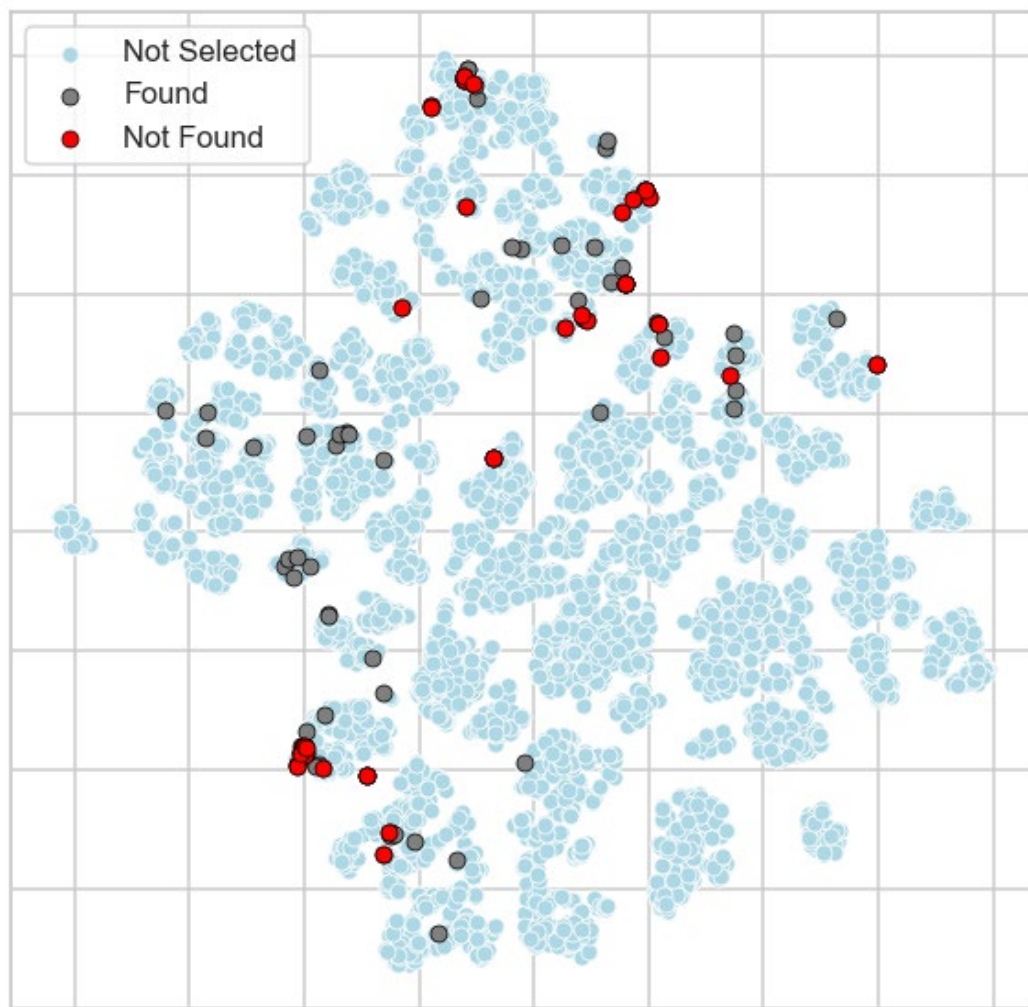
Cycle 3 - 37 of the Top 100 Molecules Found



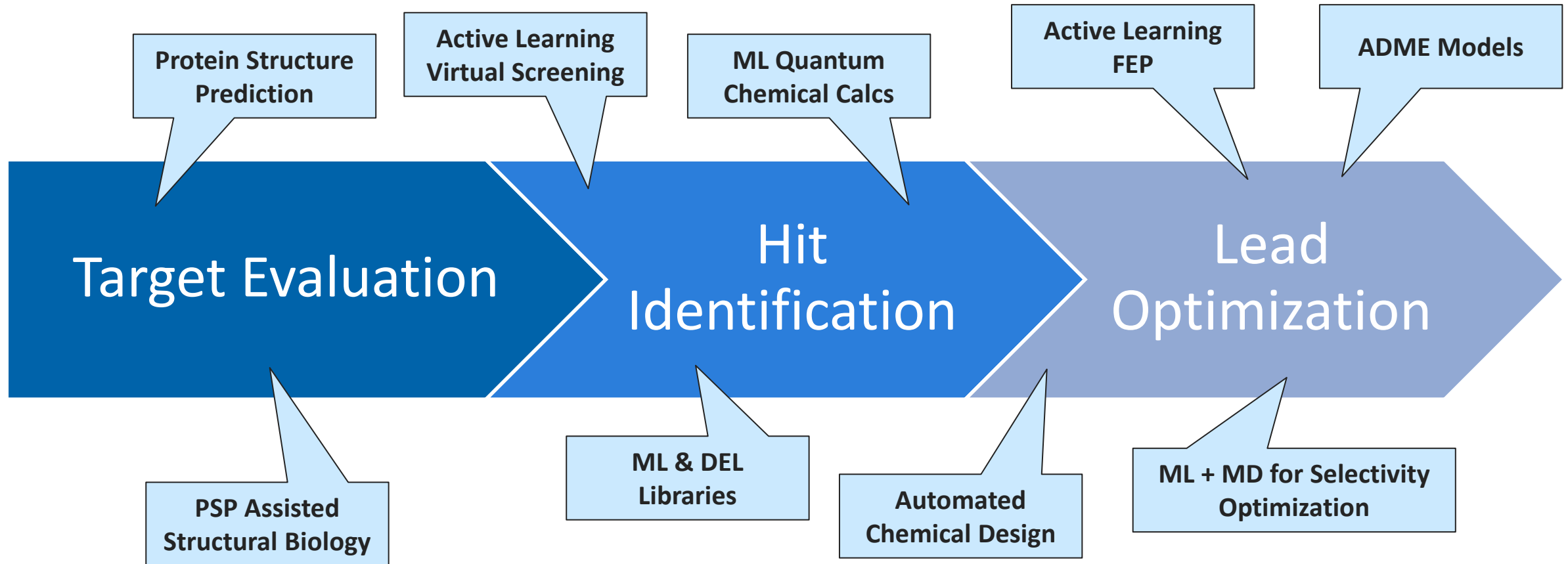
Sampled 2%



Comparing Top 100 Compounds Found and Not Found in 300 Samples (3%)



ML Has Impact Across the Drug Discovery Process



ML is One Component in a Collaboration Between Experiment and Computation

